

**DESIGN AND DEVELOPMENT OF A SELF-BALANCING MOBILE
ROBOT WITH A COLLINEAR FOUR-WHEEL MECANUM DRIVE**

A Thesis

Presented to

the Faculty of California Polytechnic State University,

San Luis Obispo

In Partial Fulfillment

of the Requirements for the Degree

<AUTHOR INPUT REQUIRED: Official degree title and program title>

by

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<AUTHOR INPUT REQUIRED: Publication deadline month and year>

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ABSTRACT

Design and Development of a Self-Balancing Mobile Robot with a Collinear Four-Wheel Mecanum Drive

Ahkar Kyaw

Dynamically balanced mobile robots combine a compact support polygon with active stabilization, but conventional two-wheel designs do not provide direct lateral motion. This thesis documents the design and development of Zephyr, a self-balancing mobile robot whose four Mecanum wheels are arranged along one common wheel-axis line rather than at the corners of a rectangular chassis. The project integrates four CubeMars AK40-10 actuators, an ISM330DHCX inertial measurement unit, a controller-area-network actuator bus, radio control, a local display, a six-cell lithium-polymer power source, embedded real-time firmware, and a developing custom printed-circuit board. A geometry-parameterized Mecanum-wheel Jacobian is derived so the model remains valid until the physical wheel order, roller handedness, and axial locations are measured. The balancing dynamics are represented by a nonlinear wheeled inverted-pendulum model and a linearized upright model suitable for state-feedback design. The current firmware demonstrates sensor acquisition, attitude-estimation infrastructure, radio-command decoding, display operation, actuator communication, feedback monitoring, and safety-gated command flow on an ESP32 prototype; the final balance controller, wheel-torque allocation, STM32 migration, and complete experimental validation remain in progress. Validation is therefore specified through reproducible simulation and hardware procedures rather than claimed as completed. <INSERT RESULT: Report final RMS pitch error, maximum pitch excursion, settling time, overshoot, steady-state error, disturbance-recovery limit, planar-velocity tracking error, control-loop timing, and CAN error statistics after testing.> The principal contribution of the present work is a

traceable mechatronic architecture and analysis framework for evaluating whether a collinear four-wheel Mecanum arrangement can combine upright stabilization with controllable longitudinal, lateral, and yaw motion. Quantitative performance conclusions remain pending the measurements identified in this draft.

Keywords: self-balancing robot, Mecanum wheel, omnidirectional mobility, wheeled inverted pendulum, embedded control, CAN bus, mechatronics

ACKNOWLEDGMENTS

<AUTHOR INPUT REQUIRED: Acknowledge the thesis advisor, committee members, laboratory support, funding or equipment support, collaborators, friends, and family as appropriate. Delete this page if acknowledgments are not included.>

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LIST OF SYMBOLS

Symbol	Definition and SI unit
A, B, C, D	State-space matrices
b_x, b_θ	Equivalent viscous damping coefficients, N·s/m and N·m·s/rad
E	Reduced mass matrix
F_x, F_y	Body-frame generalized forces, N
g	Gravitational acceleration, m/s ²
H	Wheel-to-body kinematic matrix
I_p	Body pitch moment of inertia about its center of mass, kg·m ²
K	Feedback-gain matrix
ℓ	Distance from the balancing axis to the body center of mass, m
m_b	Body mass above the wheel axis, kg
m_t	Equivalent translated mass, kg
M_z	Yaw moment, N·m
q	Generalized-coordinate vector
r	Effective wheel radius, m
s_i	Roller-orientation sign of wheel i , dimensionless
u	Control input
v_x, v_y	Body-frame longitudinal and lateral velocities, m/s
W	Positive-definite actuator-allocation weighting matrix
x, y	Inertial-frame planar position, m
y_i	Axial position of wheel i in the body frame, m
α	Complementary-filter coefficient, dimensionless
θ	Pitch angle from the upright equilibrium, rad
τ_i	Wheel- i actuator torque, N·m
ψ	Yaw angle, rad
ω_i	Wheel- i angular velocity, rad/s
ν	Body twist $[v_x, v_y, \dot{\psi}]^T$

LIST OF ABBREVIATIONS, ACRONYMS, AND NOMENCLATURE

Term	Definition
ADC	Analog-to-digital converter
BLDC	Brushless direct-current
BMS	Battery-management system
CAN	Controller area network
COM	Center of mass
CRSF	Crossfire serial protocol
DMA	Direct memory access
DOF	Degree of freedom
E-stop	Emergency stop
FOC	Field-oriented control
IMU	Inertial measurement unit
LQR	Linear-quadratic regulator
MCU	Microcontroller unit
PCB	Printed circuit board
PID	Proportional-integral-derivative
RMS	Root mean square
RTOS	Real-time operating system
SPI	Serial peripheral interface
TWAI	Two-wire automotive interface, Espressif's CAN controller
UART	Universal asynchronous receiver-transmitter
WIP	Wheeled inverted pendulum

Chapter 1

INTRODUCTION

1.1 Motivation

Mobile robots usually obtain static stability from a support polygon formed by three or more ground contacts. A dynamically balanced robot accepts a narrower support region and maintains the body near an unstable upright equilibrium by continuously commanding its wheels. This arrangement can reduce the platform width and enable rapid changes in the direction of travel, but it also makes stabilization a prerequisite for every other task. Sensor bias, computation delay, actuator saturation, gearbox compliance, wheel slip, and power interruption can therefore affect both mobility and safety. The wheeled inverted pendulum is a standard model for studying this class of system because it exposes the coupling between translation and body pitch while remaining tractable for control design [1], [2].

A conventional two-wheel balancing robot can move longitudinally and turn about a vertical axis, but it cannot command lateral translation without first changing heading. Mecanum wheels introduce passive rollers whose axes are inclined relative to the wheel plane. With suitable roller orientations and independent wheel actuation, the roller constraints redirect wheel forces so that a mobile base can produce longitudinal, lateral, and yaw motion [3–8]. Most four-wheel Mecanum platforms place the wheels at the corners of a rectangular chassis and rely on a broad, statically stable footprint. Zephyr instead places four Mecanum wheels on one common wheel-axis line and uses active pitch stabilization. The concept therefore combines the control problem of a wheeled inverted pendulum with the kinematic allocation problem of a redundant omnidirectional drive.

The collinear arrangement is not treated in this thesis as a rectangular Mecanum base

with a vanishing wheelbase. The wheel-center positions, roller signs, and balance coordinate are defined explicitly, and the mobility of the actual arrangement is determined from the rank of its geometry matrix. This treatment is necessary because lateral and yaw authority depend on the measured axial locations and roller handedness. An incorrect sign convention could produce an apparently plausible model that commands the wrong combination of wheels.

<INSERT FIGURE 1.1: Create an annotated orthographic or isometric view of Zephyr showing all four wheel centers on the common axis, wheel numbers 1 through 4, left- and right-handed rollers, body x-y-z axes, positive pitch and yaw directions, center of mass, and the plane of ground contact>
 What the figure should show: Create an annotated orthographic or isometric view of Zephyr showing all four wheel centers on the common axis, wheel numbers 1 through 4, left- and right-handed rollers, body x-y-z axes, positive pitch and yaw directions, center of mass, and the plane of ground contact
 Likely source: final SolidWorks assembly or a labeled photograph of the prototype

Figure 1.1. Zephyr robot concept and principal degrees of freedom.

1.2 Engineering Problem

The engineering problem is to determine whether a compact robot with four collinear Mecanum wheels can be designed and implemented so that it (1) maintains an upright body, (2) produces controllable motion in the horizontal plane, and (3) remains electrically and computationally safe when communication, sensing, or power faults occur. Solving this problem requires more than selecting a balance gain. The mechanical structure defines the center-of-mass height and pitch inertia; the roller arrangement defines the attainable planar wrench; the actuators introduce torque, speed, backlash, and communication limits; the IMU and estimator define the observable balance state; and the embedded schedule determines how much delay and jitter enter the feedback loop.

The project repository identifies Zephyr as a collinear Mecanum-drive self-balancing robot [9]. The current evidence shows an ESP32-based bring-up platform, an ISM330DHCX IMU interface, CRSF radio reception, an ST7789 display interface, CubeMars actuator com-

munication over a 1 Mbit/s CAN bus, feedback and fault monitoring, and a safety-gated motor-command path. The four-actuator data structures exist, but the inspected snapshot enabled only a partial actuator configuration and did not contain a completed balance-control algorithm, verified wheel-torque allocation, or hardware performance data. A NUCLEO-H723ZG/STM32H723ZG implementation exists as a scaffold or migration target rather than the demonstrated primary platform. The KiCad design is also a developing design rather than a manufactured and validated controller board.

<VERIFY FROM REPOSITORY: Reinspect the current main branch and all additional branches after exporting the repository as a ZIP. Confirm the commit hash used for the final thesis, the exact contents of 03_electrical, 04_firmware, 05_simulation, and 99_archive, and whether any balance-controller, wheel-allocation, CAD, test-data, or PCB-routing files were added after commit 97e5a0a.>

1.3 Research Questions and Objectives

The thesis is organized around questions that can be answered by design evidence, analysis, and repeatable testing. The questions avoid a claim of novelty based solely on using Mecanum wheels; they focus on the consequences of combining collinear geometry with dynamic stabilization.

Table 1.1. Research questions, objectives, and verification evidence

Research question	Engineering objective	Required verification evidence
RQ1. What planar motions are kinematically achievable with the measured collinear wheel geometry and roller pattern?	Derive the geometry matrix, determine its rank, and establish wheel-speed and wrench-allocation mappings.	Measured wheel coordinates and roller signs; analytical rank test; numerical inverse/forward kinematic tests.
RQ2. Can the completed robot maintain upright balance within defined limits while stationary and while receiving motion commands?	Implement a discrete balance controller with verified state estimation, saturation, and fault handling.	Pitch, pitch-rate, command, current, and timing logs from repeated balance and motion tests.
RQ3. How do actuator limits, roller slip, delay, and mass-property uncertainty affect performance?	Develop symbolic and numerical models and perform sensitivity analysis before hardware tuning.	Parameter record, simulation cases, model-to-hardware comparison, and uncertainty bounds.
RQ4. Does the embedded and electrical architecture support safe, maintainable experimentation?	Separate portable logic from hardware code, enforce command ownership, and verify power and CAN integrity.	Code-path review, task-timing measurements, power checks, CAN error statistics, and fault-injection results.

Note. Quantitative acceptance thresholds remain proposed until the author and advisor approve final performance requirements.

The primary objective is to produce an integrated research platform and a defensible method for evaluating it. Supporting objectives are to document the mechanical and electrical interfaces, derive a model that respects the nonstandard wheel arrangement, implement a modular real-time firmware architecture, establish a safe integration procedure, and quantify balance and planar-motion performance. The thesis does not assume that all objectives have already been achieved. Each result is assigned a status and is linked to evidence or an explicit placeholder.

1.4 Scope, Constraints, and Contributions

The scope includes mechanical arrangement, embedded electronics, actuator communication, state estimation, kinematics, reduced dynamics, control design, simulation, integration, and laboratory validation. Autonomous navigation, perception, mapping, and application-specific payloads are outside the present scope. The model treats contact with a level rigid surface and initially represents each Mecanum wheel by an idealized roller constraint. Detailed roller compliance and contact-force variation are reserved for model refinement because published work shows that individual roller contact and slip can create motion errors that an ideal kinematic model does not capture [10]–[11].

Several constraints materially shape the design. The four actuators are integrated CubeMars AK40-10 units with internal field-oriented control, a 10:1 gearbox, a single internal encoder, and CAN communication [12]. Their backlash and internal control mode affect low-speed reversals. The power source is a six-cell lithium-polymer battery, for which the fully charged voltage must be used when rating converters and protection components. The current prototype uses an ESP32, while the STM32H723ZG remains a planned higher-performance target. The final wheel order, roller pattern, center of mass, inertias, controller gains, current limits, and sampling rates have not yet been documented with sufficient evidence for a final thesis and are therefore represented by explicit verification items.

The supported contribution at this stage is an integrated design framework rather than a completed performance claim. The framework combines: a geometry-parameterized Jacobian for four collinear Mecanum wheels; a reduced nonlinear and linear balance model; a proposed allocation and control structure; a modular firmware architecture with safety-owned actuator permission; an electrical design review tied to the active firmware pin contract; and a test matrix that converts remaining work into measurable thesis evidence. The originality of the physical arrangement may be discussed only after a broader prior-art search and advisor review.

<INSERT FIGURE 1.2: Show the iterative workflow linking requirements, CAD and mass properties, schematic and PCB design, symbolic modeling, controller simulation, firmware bring-up, current-limited bench tests, restrained balancing, free balancing, and model-to-hardware comparison>

What the figure should show: Show the iterative workflow linking requirements, CAD and mass properties, schematic and PCB design, symbolic modeling, controller simulation, firmware bring-up, current-limited bench tests, restrained balancing, free balancing, and model-to-hardware comparison

Likely source: author-created vector diagram based on the documented thesis process

Figure 1.2. Research and development workflow used for the Zephyr project.

1.5 Thesis Organization

Chapter 2 reviews wheeled inverted pendulums, omnidirectional mobility, Mecanum-wheel modeling, state estimation, control, and embedded implementation. Chapter 3 converts the research questions into system requirements and interfaces. Chapters 4 and 5 document the mechanical and electrical designs. Chapter 6 derives the kinematic and dynamic models, and Chapter 7 defines the control and estimation strategy. Chapter 8 describes the firmware architecture and its demonstrated status. Chapter 9 specifies the simulation and analysis workflow. Chapter 10 presents completed and proposed integration procedures, while Chapter 11 provides ready-to-fill results tables and a cautious discussion of the evidence available at this draft stage. Chapter 12 summarizes supported conclusions and remaining work. The appendices preserve parameter, wiring, configuration, test, evidence, and compliance records needed to complete the final thesis.

Chapter 2

BACKGROUND AND LITERATURE REVIEW

2.1 Dynamically Balanced Mobile Robots

2.1.1 Wheeled inverted-pendulum dynamics

A self-balancing wheeled robot is commonly modeled as an inverted pendulum whose pivot is accelerated by the driven wheels. The upright configuration is open-loop unstable: a small pitch displacement produces a gravitational moment that increases the displacement unless wheel acceleration moves the contact region beneath the center of mass. Grasser et al. demonstrated this coupling in JOE, a two-wheel mobile inverted pendulum whose mechanical, power-electronic, and control subsystems were designed together [1]. Pathak, Franch, and Agrawal subsequently used partial feedback linearization to separate velocity and position objectives from the internal balance dynamics [2]. These systems establish the central lesson for Zephyr: translation and pitch cannot be designed as independent subsystems when the same wheel torques support both.

The full equations are nonlinear because the body center of mass moves on an arc about the wheel axis and because motor reaction torque acts on the body. Linear control is nevertheless useful near the upright equilibrium. A linearized state-space model permits controllability analysis, pole placement, and linear-quadratic regulation. LQR selects a state-feedback gain by minimizing a quadratic measure of state deviation and control effort [13], while nonlinear analysis is still required to understand the finite region in which the linear approximation is credible [14]. Recent comparative work continues to use LQR as a transparent baseline for inverted-pendulum stabilization [15]. For Zephyr, a linear controller is therefore appropriate as an initial implementation if the nonlinear plant, actuator saturation, and fall thresholds remain part of the validation.

2.1.2 Underactuation, nonminimum-phase behavior, and practical limits

A wheel torque both accelerates the base and applies an equal and opposite reaction torque to the body. The balance coordinate is not independently actuated, which makes the robot underactuated. A commanded translation generally requires a transient body lean, and an aggressive position loop can conflict with the inner balance loop. This behavior motivates a cascade in which pitch stabilization has the highest bandwidth, forward velocity or position is slower, and yaw and lateral objectives are allocated without consuming the torque reserve required for balance. The final bandwidth separation must be based on measured delay and actuator dynamics rather than a generic ratio.

Practical limits define the recoverable set. Finite motor current, wheel-ground friction, gearbox backlash, battery voltage, sensor range, and computation delay limit the pitch angle and angular rate from which the robot can recover. A controller that is stable in an

unconstrained model may fail after a command saturates. Accordingly, the Zephyr model includes torque and rate limits, and the proposed tests record both pitch response and motor effort. No recovery envelope is claimed until the physical robot is tested under controlled conditions.

2.2 Omnidirectional and Mecanum-Wheel Mobility

2.2.1 Mecanum-wheel principle

Hon's wheel patent describes a load-bearing wheel with peripheral rollers arranged obliquely to the wheel plane [4]. Rotation of the main wheel creates a contact velocity that can be decomposed into a constrained component and a roller-supported component. Combining wheels of opposite handedness enables a platform to synthesize motion directions that a conventional fixed wheel cannot. Pin and Killough placed this concept within the broader family of holonomic and omnidirectional platforms [3], while Ferrière, Raucent, and Campion examined the geometry and design of omnidirectional wheels [7]. Gferrer later provided a geometric treatment of the Mecanum wheel itself [5].

The ideal kinematic model treats each roller as imposing a linear velocity constraint. This abstraction is valuable for deriving a Jacobian and checking controllability, but it does not fully represent discrete roller contact, varying normal load, deformation, or slip. Bayar and Ozturk showed that roller contact forces affect Mecanum-wheel behavior [10]. More recent experimental modeling work likewise reports the need to consider contact and slip when high accuracy is required [16], [11]. Zephyr is especially sensitive to these effects because the wheel forces must stabilize pitch as well as produce planar motion.

2.2.2 Mobility classification and geometry rank

Campion, Bastin, and D'Andréa-Noel classified wheeled-mobile-robot models according to their kinematic constraints and mobility [17]. Muir and Neuman also showed how wheel constraints can be assembled into a systematic kinematic model [8]. These approaches emphasize that mobility follows from the constraint matrix, not from a wheel label. A four-wheel Mecanum platform is not automatically holonomic if the roller signs or locations make the rows of its Jacobian dependent.

Indiveri derived kinematic and control relationships for Swedish-wheel omnidirectional robots [6]. Conventional four-corner formulas use two geometric lever arms and a prescribed X or O roller arrangement. They cannot be copied directly to Zephyr because all wheel centers lie on one line. For a collinear arrangement, yaw arises from longitudinal force differences across the axial wheel locations, while lateral force arises from the roller signs. Full planar mobility requires both roller handednesses and a distribution of wheel locations that keeps the longitudinal, lateral, and yaw columns independent. Chapter 6 expresses this requirement as a rank condition that can be evaluated after measurement.

<INSERT FIGURE 2.1: Create a synthesis diagram showing how inverted-pendulum dynamics, Mecanum-wheel constraints, IMU state estimation, actuator/CAN limits, real-time scheduling, and safety supervision interact in Zephyr>
 What the figure should show: Create a synthesis diagram showing how inverted-pendulum dynamics, Mecanum-wheel constraints, IMU state estimation, actuator/CAN limits, real-time scheduling, and safety supervision interact in Zephyr
 Likely source: author-created vector diagram based on references [1] through [18] and the repository architecture

Figure 2.1. Relationship among balancing, omnidirectional mobility, state estimation, and embedded implementation.

2.3 Control and State Estimation

2.3.1 PID, state feedback, and LQR

PID control remains attractive because it is easy to implement and tune from measured responses. A pitch-angle loop can command wheel torque from angle, rate, and integral error, while an outer velocity loop shifts the pitch reference. The method becomes harder to reason about when translation, pitch, yaw, lateral motion, actuator saturation, and allocation interact. Anti-windup is essential whenever the command is limited.

State feedback provides a direct way to use pitch, pitch rate, translational velocity, and position in one control law. LQR supplies a repeatable gain-selection framework once the linear model and weighting matrices are documented [19], [13]. It does not remove the need for engineering judgment: state scaling changes the meaning of the weights, unmodeled delay can erode phase margin, and actuator limits invalidate the unconstrained optimum. The Zephyr simulation draft uses LQR as a proposed design method, but no gain is described as implemented until it appears in firmware and is verified on hardware.

2.3.2 Inertial sensing and complementary filtering

An accelerometer provides a gravity-referenced attitude estimate when translational acceleration is small, whereas a gyroscope measures angular rate with high short-term fidelity but accumulates bias when integrated. A complementary filter combines the low-frequency accelerometer reference with the high-frequency integrated gyroscope signal. Mahony, Hamel, and Pflimlin placed complementary attitude estimation on a rigorous nonlinear foundation [20]. The current Zephyr code path includes an IMU driver, mounting transform, gyro calibration, an attitude estimator, and an IMU state snapshot. The physical mounting transform and sign convention must be verified by controlled rotations before any controller is armed.

2.4 Embedded Actuation and Communication

2.4.1 Integrated actuators and local motor control

Zephyr uses four CubeMars AK40-10 actuators. The manufacturer describes the unit as a brushless motor, 10:1 planetary gearbox, internal driver, and magnetic encoder integrated

into one module [12]. Published ratings include 24 V nominal voltage, 1.3 N·m rated torque, 4.1 N·m peak torque, 370 r/min rated speed, 435 r/min no-load speed, 2.7 A rated current, and 7.3 A peak current. These are component ratings, not selected operating limits. The installed firmware version, operating mode, current ceiling, speed limits, and protection settings require hardware confirmation.

An integrated actuator moves the inner current and field-oriented-control loops out of the central MCU, reducing the required update rate of the supervisory controller. It also hides some motor dynamics and introduces protocol, firmware, and mode-dependent behavior. For initial simulation, the actuator can be represented as a saturated command path with a measured delay or first-order response. A higher-order electrical motor model is justified only if current-loop data and parameters are available. Gearbox backlash is relevant because balance commands reverse near zero speed, where the wheel may not immediately follow the internal motor encoder.

2.4.2 CAN networks and real-time scheduling

CAN provides prioritized, frame-based communication and automatic error detection. The ESP32 exposes a Classical CAN-compatible controller named TWAI, but it still requires an external physical-layer transceiver [21], [22]. The current prototype uses an Adafruit CAN Pal based on the TJA1051T/3 transceiver [23], [24]. A linear bus with short stubs and one 120 Ω termination at each physical end is the baseline topology; TI's selectable-termination reference design explains why the distal terminations are required [25]. Bus bitrate alone does not establish real-time performance. Frame identifiers, payloads, feedback periods, arbitration, timeouts, and error counters must be logged under load.

Real-time software must complete sensing, estimation, control, and command transmission before their deadlines. Liu and Layland established foundational schedulability results for periodic fixed-priority tasks [18]. Zephyr uses FreeRTOS concepts, but its final schedule should be verified by measurement because driver blocking, serial bursts, display updates, logging, and CAN retries can create execution-time variability [26]. The safety architecture should treat stale data and missed deadlines as faults rather than allowing the last command to persist indefinitely.

2.5 Research Gap and Design Implications

The literature separately provides mature methods for wheeled inverted pendulums and for statically stable omnidirectional platforms. The Zephyr concept places those methods in direct interaction. Its four wheels add actuation redundancy, but the collinear geometry removes the conventional rectangular footprint and changes the yaw moment arm. Its Mecanum rollers provide lateral authority, but roller slip perturbs both odometry and the force available for stabilization. Its integrated motor controllers reduce central computation, but CAN delay and gearbox compliance become part of the balance loop.

The practical gap addressed by this thesis is therefore a traceable system design and validation method for a collinear four-wheel Mecanum balancing robot. The work does not

claim that no related arrangement exists. Instead, it asks whether the actual geometry has full planar rank, whether the built platform can stabilize under its measured limits, and how closely a reduced model predicts that behavior. This framing leads directly to the requirements and evidence matrix in Chapter 3.

Table 2.1. Comparison of relevant mobile-robot architectures and methods

System or method	Established contribution	Limitation relative to Zephyr	Design implication
Two-wheel inverted pendulum [1], [2]	Coupled translation and pitch dynamics; balance and position control	No direct lateral-motion allocation	Preserve a high-bandwidth balance objective while adding redundant wheel allocation.
Four-corner Mecanum platform [3], [5]–[11]	Planar holonomic kinematics and omnidirectional motion	Assumes a statically stable rectangular wheel geometry	Derive the Jacobian from measured collinear wheel locations and roller signs.
Complementary attitude estimation [20]	Combines gyro short-term response with gravity reference	Acceleration and mounting errors can bias pitch	Calibrate bias and axes; validate under commanded acceleration.
LQR/state feedback [19], [13], [15]	Systematic multistate feedback near an equilibrium	Depends on model, state scaling, and unsaturated operation	Document weights, discretization, limits, and robustness tests.
CAN-based integrated actuators [22], [12], [23]	Local FOC and compact wiring	Network delay, mode settings, encoder location, and backlash	Measure timing and verify every installed actuator parameter.

Chapter 3

SYSTEM REQUIREMENTS AND ARCHITECTURE

3.1 Requirements Development

Requirements translate the research questions into observable evidence. The repository does not contain an approved numerical specification for pitch error, speed, runtime, or thermal margin. Numerical thresholds in Table 3.1 are therefore identified as author- or advisor-controlled values rather than silently selected targets. A requirement becomes verified only when its method, instrumentation, operating condition, and result are recorded.

Table 3.1. System requirements and verification matrix

ID	Requirement	Acceptance criterion	Verification method	Current status
FR-01	The robot shall command four independently addressable actuators.	All four installed node IDs respond and report valid state.	CAN enumeration and command/feedback test.	Partial: four slots exist; only the verified active IDs may be claimed.
FR-02	The system shall estimate body pitch and pitch rate.	Correct sign and scale through the required angle/rate range; freshness monitored.	Static orientations, controlled rotations, and timestamp audit.	Partial: acquisition and estimator infrastructure exist; final calibration pending.
FR-03	The completed controller shall maintain upright balance.	<AUTHOR INPUT REQUIRED: RMS pitch, peak excursion, settling time, and test duration limits>.	Restrained then free-standing repeat trials.	Not demonstrated in available evidence.
FR-04	The wheel arrangement shall provide commanded x, y, and yaw motion while upright.	Measured geometry matrix rank equals 3 and commanded signs match measured motion.	Analytical rank test, wheel-direction test, velocity trials.	Pending geometry and balance validation.
PR-01	The real-time loop shall meet its deadline.	<AUTHOR INPUT REQUIRED: period, maximum execution time, and jitter limit>.	Timestamp or logic-analyzer measurement under worst-case load.	Not measured.
PR-02	CAN communication shall remain within the allowed age and error limits.	<AUTHOR INPUT REQUIRED: feedback age, timeout, bus-off, and frame-loss limits>.	Bus logging and fault injection.	Basic motor communication demonstrated; statistics absent.
ER-01	Every device shall operate within its rated voltage range.	Measured rail min/max within component limits during startup, idle, and peak load.	DMM and oscilloscope measurements at load.	Design review in progress; final PCB unverified.
ER-02	Power conductors and protection shall support the selected actuator limits.	Temperature rise and voltage drop remain below approved limits; faults clear safely.	Current-limited test, load test, thermal imaging, fuse test.	PCB and protection implementation pending.
SR-01	Loss of valid command, IMU state, or actuator feedback shall remove torque permission.	Command becomes zero/disabled within <AUTHOR INPUT REQUIRED: timeout>.	Unplug, stale-data, and injected-fault tests.	Safety-gated architecture reported; final validation pending.
MR-01	Firmware shall separate portable logic from platform-specific I/O.	Common modules compile independently and platform code owns hardware access.	Repository structure and build review.	Implemented in current architecture at a structural level.

Note. FR = functional, PR = performance, ER = electrical, SR = safety, and MR = maintainability requirement.

3.2 System Architecture

The system is divided into mechanical, power, sensing, computation, communication, user-interface, and safety subsystems. The mechanical structure carries the battery, electronics, and body mass above the common wheel axis. Four independent actuators connect the controller to the ground through the Mecanum wheels. The IMU measures body motion; the radio receiver provides operator commands; the display provides local status; and CAN exchanges commands and feedback with the actuators.

Power enters from the six-cell battery. The actuator branches use the battery domain directly, subject to fusing, switching, and protection that must be confirmed in the final schematic. A buck converter produces the logic rail, and local regulation or the controller-board regulator produces 3.3 V where required. Logic ground and power return share a controlled reference at the system level, but high-current motor return paths must not force switching current through sensitive sensor or transceiver return paths.

<INSERT FIGURE 3.1: Draw a system block diagram containing the 6S battery, master disconnect and emergency stop, branch fuses, optional BMS/protection board, actuator power branches, logic buck converter, ESP32 prototype and STM32 target, ISM330DHCX IMU, CAN transceiver, four AK40-10 actuators, RadioMaster receiver, ST7789 display, and debug/logging interface>

What the figure should show: Draw a system block diagram containing the 6S battery, master disconnect and emergency stop, branch fuses, optional BMS/protection board, actuator power branches, logic buck converter, ESP32 prototype and STM32 target, ISM330DHCX IMU, CAN transceiver, four AK40-10 actuators, RadioMaster receiver, ST7789 display, and debug/logging interface

Likely source: KiCad schematic, firmware architecture, and final hardware inventory

Figure 3.1. System-level architecture of the Zephyr robot.

Information flows in the opposite direction from control authority. Drivers convert peripheral data into timestamped measurements. Estimation converts calibrated IMU data into pitch and rate. A supervisor decides whether the current operating mode permits torque. The controller produces a desired body wrench or wheel command. An allocation stage maps the body command to four actuators. The motor task transmits only commands that have passed safety checks. Feedback and faults return through CAN and update the supervisor. This ownership prevents the radio, UI, or a bring-up test from bypassing the safety decision in production operation.

<INSERT FIGURE 3.2: Use separate arrow styles for battery power, regulated power, raw sensor data, estimated state, operator command, safety permission, controller output, CAN command, CAN feedback, faults, and telemetry>
What the figure should show: Use separate arrow styles for battery power, regulated power, raw sensor data, estimated state, operator command, safety permission, controller output, CAN command, CAN feedback, faults, and telemetry
Likely source: firmware module diagram and electrical interface diagram

Figure 3.2. Power, sensing, command, and feedback paths among major subsystems.

3.3 Subsystem Interfaces

Table 3.2. Major subsystem interfaces

Producer	Consumer	Interface	Information or power	Required verification
6S battery/protection	Four actuator branches	VBAT and power return	High-current energy	Polarity, fuse rating, conductor loss, peak droop, disconnect behavior.
Logic converter	Controller and peripherals	5 V and 3.3 V rails	Regulated power	Startup, ripple, load transient, thermal rise, undervoltage.
ISM330DHCX	MCU	SPI plus interrupt lines	Acceleration, angular rate, data-ready	WHO_AM_I, rate, range, axes, timestamp, missed samples.
RadioMaster RP2 V2	MCU	CRSF UART	Operator channels and link state	Baud rate, TX/RX crossing, failsafe, frame timeout.
MCU	ST7789 display	SPI and control GPIO	Status and diagnostics	Voltage, pin mapping, refresh load, no control-loop blocking.
MCU/TWAI	TJA1051T/3	TXD/RXD	CAN logic	3.3 V compatibility, standby state, error behavior.
CAN transceiver	AK40-10 network	CANH/CANL and reference	Commands, feedback, faults	Linear topology, two terminations, bitrate, IDs, timeouts.
Estimator Supervisor/safety	Supervisor/control Motor command path	Timestamped snapshot Mode and torque permission	Pitch, rates, validity and age Authorization and zero-torque request	Atomic access, freshness, calibration status. Priority, deterministic fault response, no bypass.

3.4 Design Tradeoffs

Four actuators provide redundancy and independent force allocation, but they increase current demand, CAN traffic, wiring, configuration effort, and the number of possible sign errors. Mecanum wheels provide lateral authority, but their rollers introduce compliance, vibration, changing contact conditions, and slip. Integrated actuators reduce the burden on the central MCU, but their internal loop settings and motor-side encoder cannot be treated as an ideal wheel-torque source. A custom PCB can improve wiring and grounding, but it creates manufacturing risk if the schematic is released before the firmware pin contract and power-tree design are frozen.

The ESP32 is suitable for rapid bring-up because the current firmware and peripheral interfaces are already developed on that platform. The STM32H723ZG offers greater deterministic-control capability and FDCAN resources [27], but migration creates a second implementation that must reproduce the same common logic and safety behavior. The selected strategy is to stabilize architecture and interfaces in portable modules, complete hardware evidence on one platform, and migrate only when the control and timing requirements justify the change.

3.5 Architecture Status

The repository evidence supports a modular architecture and several functioning device paths, but not a completed system. The mechanical assembly is reported to be approximately 80 percent complete and the motors have been moved from the prototype firmware. The electrical design has progressed from point-to-point wiring toward a custom board, but the reviewed PCB state was unrouted or pre-fabrication. The firmware contains sensing, communication, UI, supervisor, and safety infrastructure, but the final balance law and wheel allocation remain absent or unverified. The simulation work provides a symbolic and numerical starting point, not experimentally validated results. These status distinctions govern the wording of the remaining chapters.

Chapter 4

MECHANICAL DESIGN

4.1 Mechanical Concept

Zephyr is organized about a common horizontal wheel axis. Four independently driven Mecanum wheels are distributed along this line, and the body mass is carried above the line so that gravity produces an unstable pitch moment. In the nominal upright configuration, longitudinal motion is perpendicular to the common wheel axis, lateral motion is parallel to that axis, yaw is rotation about the vertical axis, and pitch is rotation about the common wheel axis. This definition must be reconciled with the final CAD coordinate system and IMU mounting axes before signs are frozen.

A rectangular four-corner Mecanum chassis obtains yaw authority from longitudinal and lateral offsets. In the collinear design, all wheel-center longitudinal coordinates are nominally zero. Yaw authority must therefore come from the differences in longitudinal force applied at distinct axial positions. Lateral force depends on the roller handedness. The physical order of left- and right-handed wheels is consequently a functional design variable rather than a cosmetic assembly detail.

<INSERT FIGURE 4.1: Insert a high-resolution photograph of the current mechanical prototype on a plain background. Label wheel numbers, actuator bodies, structural frame, battery location, electronics enclosure, IMU location, cable paths, and balance axis. Identify which components are temporary>
What the figure should show: Insert a high-resolution photograph of the current mechanical prototype on a plain background. Label wheel numbers, actuator bodies, structural frame, battery location, electronics enclosure, IMU location, cable paths, and balance axis. Identify which components are temporary
Likely source: physical prototype photograph

Figure 4.1. Current Zephyr mechanical prototype with major components identified.

<INSERT FIGURE 4.2: Insert an exploded CAD view with a bill-of-material callout for frame plates, printed parts, actuator mounts, wheel hubs, bearings or supports, fasteners, battery retention, electronics mounts, covers, and cable restraints>
What the figure should show: Insert an exploded CAD view with a bill-of-material callout for frame plates, printed parts, actuator mounts, wheel hubs, bearings or supports, fasteners, battery retention, electronics mounts, covers, and cable restraints
Likely source: final SolidWorks assembly and drawing package

Figure 4.2. Exploded CAD view of the chassis, actuator mounts, wheels, and electronics structure.

4.2 Coordinate Frames and Wheel Arrangement

The body frame $\{B\}$ is placed on the common wheel-center line at a defined axial origin. The x_B axis points in the nominal forward direction, y_B follows the common wheel axis, and z_B points upward when the robot is upright. Wheel i is located at $p_i = [0, y_i, 0]^T$ for the ideal collinear model. A roller sign s_i is assigned from the direction in which positive wheel rotation couples to positive lateral velocity. The sign must be determined by inspection and verified by lifting the platform and commanding one wheel at a time.

The balance angle θ is zero when the designated body reference plane is vertical. Positive pitch, yaw, wheel rotation, motor current, and torque directions must be documented on the same drawing. The IMU mounting rotation is then a fixed transform from the sensor frame to $\{B\}$; software should apply this transform before the estimator. Using one signed convention throughout CAD, wiring, firmware, simulation, and plots prevents the common error of compensating for a reversed motor in multiple layers.

<INSERT FIGURE 4.3: Provide a dimensioned coordinate-frame drawing with the origin, x_B - y_B - z_B directions, inertial frame, positive pitch and yaw, wheel centers y_1 through y_4 , effective wheel radius, roller angle and sign, center-of-mass coordinates, and ground plane>

What the figure should show: Provide a dimensioned coordinate-frame drawing with the origin, x_B - y_B - z_B directions, inertial frame, positive pitch and yaw, wheel centers y_1 through y_4 , effective wheel radius, roller angle and sign, center-of-mass coordinates, and ground plane

Likely source: CAD drawing supplemented by MATLAB vector graphics

Figure 4.3. Body coordinate frame, collinear wheel locations, roller signs, and balancing axis.

<AUTHOR INPUT REQUIRED: Explain why the collinear four-wheel arrangement was selected instead of a conventional two-wheel balancing platform or a statically stable four-corner Mecanum platform. Address expected lateral authority, packaging, redundancy, research value, and accepted mechanical complexity.>

4.3 Wheels and Actuators

The selected wheels are the goBILDA GripForce Mecanum Wheel Set, whose manufacturer lists a 104 mm diameter, 40A-durometer silicone rollers, 11 bearing-supported rollers per wheel, and a mass of 236 g per wheel [28]. The 52 mm nominal radius inferred from the catalog diameter is suitable for preliminary calculations, but the effective rolling radius under robot load should be measured because roller deformation and floor compliance change the kinematic scale.

Each wheel is driven by a CubeMars AK40-10 actuator. The compact integrated package reduces the number of separate motor-controller and encoder mounts. It also imposes a 10:1 planetary gearbox and a single internal encoder in the documented variant [12].

The manufacturer lists 18 arcmin of backlash. At the wheel rim, the corresponding lost motion depends on the actual output backlash and wheel radius; this value should be measured rather than inferred from the catalog alone. Backlash is particularly important near the zero-speed reversals generated by a balance controller.

<INSERT FIGURE 4.4: Show a section or exploded detail from actuator housing to wheel: actuator mounting face, output interface, hub or adapter, fasteners, wheel, any shaft support or bearing, axial retention, locating features, and torque reaction into the frame. Include critical dimensions and materials>
 What the figure should show: Show a section or exploded detail from actuator housing to wheel: actuator mounting face, output interface, hub or adapter, fasteners, wheel, any shaft support or bearing, axial retention, locating features, and torque reaction into the frame. Include critical dimensions and materials
 Likely source: manufacturing drawing and actuator/wheel CAD

Figure 4.4. Actuator-to-wheel structural interface and load path.

4.4 Chassis, Structural Interfaces, and Mass Distribution

The chassis must carry the battery and electronics rigidly relative to the IMU while transmitting wheel reaction torque among four actuator mounts. Pitch-control bandwidth can excite compliance if the body twists about the common axis or if the IMU mount flexes. The structural design should therefore use closed load paths, short actuator-mount spans, positive locating features, and fasteners sized for repeated reversals. Removable electronics and battery modules improve serviceability, but their interfaces must not introduce motion into the IMU reference.

Center-of-mass placement creates a deliberate tradeoff. A higher center of mass increases the gravitational moment for a given pitch angle and generally slows the open-loop falling motion, but it also increases the torque and energy required for recovery. A lower center of mass reduces gravitational leverage but can demand faster sensing and actuation. The final controller model requires the assembled mass, center-of-mass height and offsets, pitch inertia, and wheel/actuator inertias. CAD mass properties should be checked by physical weighing and a center-of-mass or pendulum test because fasteners, wiring, printed infill, and batteries are often modeled inaccurately.

Table 4.1. Mechanical parameters required for analysis and validation

Parameter	Symbol	Unit	Required method	Current value/status
Assembled robot mass	m	kg	Calibrated scale with final battery and wiring	<MEASURE>
Body mass above wheel axis	m_b	kg	CAD grouping and component weighing	<MEASURE>
Effective wheel radius	r	m	Loaded rollout distance over multiple revolutions	Catalog radius 0.052 m; effective value unverified
Wheel axial locations	y1...y4	m	CAD and physical measurement from the defined origin	<MEASURE>
Center-of-mass height	ℓ	m	CAD mass properties plus physical confirmation	<MEASURE>
COM x and y offsets	x_c, y_c	m	Balance/knife-edge or suspension method	<MEASURE>
Pitch moment of inertia	I_p	kg·m ²	CAD plus bifilar/trifilar or system-identification test	<MEASURE>
Yaw moment of inertia	I_z	kg·m ²	CAD plus torsional test or system identification	<MEASURE>
Wheel/rotor equivalent inertia	J_eq	kg·m ²	Manufacturer data and reflected-inertia calculation	<VERIFY>
Roller angle and signs	β , s_i	deg, —	Physical inspection and single-wheel direction test	<VERIFY FROM HARDWARE>

4.5 Manufacturability, Assembly, and Serviceability

The final drawing package should control actuator location, wheel spacing, perpendicularity between the wheel axis and body reference, and the orientation of each Mecanum wheel. Assembly should use an unambiguous wheel-number label and roller-direction mark that remains visible after installation. Threaded joints subject to reversing torque should use appropriate locking features and documented tightening practices. The battery should be mechanically retained independently of its electrical connector, and cables should include strain relief and clearance from rotating rollers.

Service access should allow the battery, controller, CAN transceiver, receiver, display, and IMU to be removed without disturbing the entire drivetrain. Connectors should be keyed or labeled so that a CAN plug cannot be inserted into a power header. Test points for logic rails, CANH, CANL, and ground should remain accessible when the robot is restrained. A removable physical stand or tether attachment is recommended for initial controller tests.

4.6 Mechanical Status and Remaining Evidence

Table 4.2. Mechanical component and interface status

Item	Evidence available	Apparent status	Information required
Four-wheel collinear concept	Project description and author statements	Selected	Dimensioned drawing and roller order
AK40-10 actuators	Repository firmware and manufacturer documentation	Motors reported moving	Installed revision, mount drawing, output interface
104 mm Mecanum wheels	Selected part and manufacturer specification	Selected/procured	Installed handedness and loaded radius
Frame/chassis	Author reports mechanical build approximately 80 percent complete	Partially manufactured	CAD, materials, dimensions, mass properties, photos
Bearings/shafts/couplings	No sufficient repository evidence available	Unconfirmed	Part numbers, fits, retention, calculations
Cable management and covers	No sufficient final evidence available	Pending or unconfirmed	Photos and assembly procedure
Structural analysis	No confirmed FEA or hand calculations	Pending	Load cases, material properties, stresses, deflections, safety factors

Chapter 5

ELECTRICAL AND ELECTRONIC DESIGN

5.1 Electrical Architecture

The electrical system has two energy domains. The high-current domain supplies the four actuators from the six-cell lithium-polymer battery. The logic domain supplies the controller, IMU, transceiver, receiver, and display through regulated rails. Separating these functions in the schematic and layout makes it possible to review conductor ratings, protection, startup behavior, and noise coupling independently while maintaining one intentional system reference.

<INSERT FIGURE 5.1: Show VBAT, protected or switched battery output, four actuator branches, logic converter input, 5 V rail, 3.3 V rail, controller, IMU, CAN transceiver, receiver, display, debug connection, and every ground-return path. Use distinct line weights for high-current and logic paths>

What the figure should show: Show VBAT, protected or switched battery output, four actuator branches, logic converter input, 5 V rail, 3.3 V rail, controller, IMU, CAN transceiver, receiver, display, debug connection, and every ground-return path. Use distinct line weights for high-current and logic paths

Likely source: final KiCad hierarchical schematic

Figure 5.1. Electrical architecture and voltage domains.

Table 5.1. Electrical components and implementation status

Function	Selected or proposed component	Evidence/status	Design issue to close
Main controller	ESP32 NodeMCU prototype	Active firmware platform	Exact board revision, regulator capacity, final PCB integration
Future controller	NUCLEO-H723ZG / STM32H723ZG	Scaffolded migration target	Migration decision, pin freeze, timing verification
IMU	ST ISM330DHCX	SPI driver and estimator path present	Final mounting, axes, supply, interrupts, calibration
CAN physical layer	Adafruit CAN Pal using TJA1051T/3	Prototype interface	Final footprint or direct transceiver, standby pin, protection, termination
Actuators	4 × CubeMars AK40-10	Motor communication/motion partially demonstrated	IDs, modes, limits, firmware, harness pinout
Radio receiver	RadioMaster RP2 V2 ExpressLRS	CRSF decoding reported	Supply, UART mapping, link-loss behavior
Display	Waveshare 2-inch ST7789 module	UI interface reported	Final 3.3 V compatibility, connector footprint, update load
5 V conversion	LT8609S design	Schematic reviewed; hardware validation unavailable	Enable threshold, layout, thermal and transient testing

Function	Selected or proposed component	Evidence/status	Design issue to close
Battery protection	BQ76942-based 6S design discussed	Recommended/separate BMS work; not confirmed manufactured	Final architecture, MOSFETs, shunt, cell connections, configuration

5.2 Battery, Protection, and Power Distribution

A six-cell lithium-polymer pack has a nominal voltage of approximately 22.2 V and a fully charged voltage of 25.2 V. Converter and protection ratings must use the maximum voltage plus credible transients, not the nominal label. The selected actuators are rated for 24 V operation [12], so the design supplies them from the battery domain rather than generating a regulated 24 V rail. The exact battery capacity, discharge rating, connector, cell condition, and allowable undervoltage threshold remain hardware verification items.

The high-current design should include a master disconnect, an emergency method that removes actuator energy, branch overcurrent protection, reverse-polarity prevention or keyed connectors, and a precharge or anti-spark strategy if connector inrush is material. A BQ76942 can monitor and protect three to ten series cells [29], but using the device requires a complete cell-tap network, current shunt, charge and discharge protection MOSFETs, thermistors, pack/load connections, configuration, and fault testing. The thesis must distinguish a separate BMS board from the main controller board if they are manufactured independently.

<INSERT FIGURE 5.2: Show the exact 6S battery connector or two connector branches, master switch or contactor, emergency stop, precharge or anti-spark element, pack fuse, per-motor or per-pair fuses, BMS/protection boundaries, current shunt, actuator connectors, LT8609S input, and high-current returns. Label ratings and connector pin numbers>
 What the figure should show: Show the exact 6S battery connector or two connector branches, master switch or contactor, emergency stop, precharge or anti-spark element, pack fuse, per-motor or per-pair fuses, BMS/protection boundaries, current shunt, actuator connectors, LT8609S input, and high-current returns. Label ratings and connector pin numbers
 Likely source: final power schematic and wiring harness drawing

Figure 5.2. Battery and high-current power-distribution schematic.

Table 5.2. Electrical power budget

Load	Quantity	Voltage	Rated/expected current	Peak/design current	Evidence
AK40-10 actuator	4	VBAT	2.7 A each manufacturer rated current	7.3 A each manufacturer peak current	Manufacturer specification; chosen limits unverified
ESP32 controller board	1	5 V input or 3.3 V rail	<MEASURE>	<MEASURE startup/transmit peak>	No final measured data
ISM330DHCX	1	3.3 V	<CALCULATE FROM SELECTED MODE>	<VERIFY>	Use exact datasheet operating mode [30]
CAN transceiver	1	3.3 V module input	<VERIFY>	<VERIFY dominant-bus load>	Adafruit/NXP documentation [23], [24]
Radio receiver	1	5 V	<MEASURE>	<MEASURE>	Confirm installed RP2 V2 revision [31]
ST7789 display module	1	3.3 V	<MEASURE at selected brightness>	<MEASURE>	Confirm module revision [32]
Logic total	—	5 V and 3.3 V	<SUM WITH MARGIN>	<LOAD-STEP TEST>	Pending

Note. Manufacturer peak ratings are not simultaneous system requirements. Final branch and pack ratings must follow the configured motor limits and measured duty cycle.

5.3 Voltage Conversion and Logic Rails

The reviewed design uses an LT8609S step-down regulator for the 5 V logic rail. The device accepts an input range that includes a fully charged 6S pack [33]. The reviewed schematic used a 2.2 μH inductor, 1 M Ω and 182 k Ω feedback resistors, a 10 pF feed-forward capacitor, a 1 μF input capacitor, and a 22 μF output capacitor, producing a nominal output near 5.03 V. Those values must be checked against the exact switching frequency, inductor current rating, capacitor bias derating, minimum load, loop stability guidance, and the manufacturer's recommended layout.

The CAN Pal, ISM330DHCX interface, and display logic were previously found connected or labeled inconsistently with their intended 3.3 V operation. The final schematic should explicitly identify the supply accepted by each installed module rather than assuming that a breakout labeled VIN is equivalent to the IC supply. The RadioMaster receiver is assigned to the 5 V rail in the current interface plan. Decoupling must be placed at each device or connector, and the 3.3 V regulator capacity must include display and transceiver transient current if those loads share the controller-board regulator.

5.4 Controller, Sensor, Receiver, and Display Interfaces

The active ESP32 pin contract inferred from the current firmware and schematic review is summarized in Table 5.3. The IMU uses SPI; the receiver uses a CRSF UART; the display uses a separate SPI-style interface; and the TWAI controller connects through TXD and RXD to the external transceiver. The table is a design baseline, not permission to fabricate. Every pin must be checked against the exact ESP32 board schematic, boot-strapping constraints, input-only pins, voltage domains, and the current repository configuration.

Table 5.3. ESP32 prototype pin assignments

Subsystem	Signal	ESP32 GPIO	Direction at MCU	Electrical note
ISM330DHCX	SCLK	18	Output	3.3 V SPI clock
ISM330DHCX	SDO/MISO	19	Input	Sensor output to MCU
ISM330DHCX	SDI/MOSI	23	Output	MCU output to sensor
ISM330DHCX	CS	27	Output	Active-low chip select
ISM330DHCX	INT1 / INT2	34 / 35	Input	Input-only GPIOs; verify pull configuration
ST7789 display	DIN / CLK	13 / 14	Output	3.3 V logic
ST7789 display	CS / DC / RST	25 / 21 / 22	Output	Use DC label, not ambiguous DS
RadioMaster RP2 V2	Receiver TX → MCU RX	16	Input	Verify crossed UART naming at connector
RadioMaster RP2 V2	MCU TX → receiver RX	17	Output	CRSF control/telemetry path
CAN transceiver	TXD	26	Output	MCU TWAI transmit to transceiver
CAN transceiver	RXD	32	Input	Transceiver receive to MCU

Note. A September schematic review contained inconsistent TX/RX naming in different descriptions. The final connector drawing must name signals by source and destination and be checked against firmware.

<INSERT FIGURE 5.3: Insert the finalized controller-interface schematic with connector pin numbers, voltage rails, decoupling, pull resistors, ESD protection, test points, and net labels matching Table 5.3. Include board/module part numbers and footprint references>
 What the figure should show: Insert the finalized controller-interface schematic with connector pin numbers, voltage rails, decoupling, pull resistors, ESD protection, test points, and net labels matching Table 5.3. Include board/module part numbers and footprint references
 Likely source: final KiCad schematic

Figure 5.3. Controller, sensor, receiver, display, and CAN interface schematic.

5.5 CAN Bus

The current firmware uses the ESP32 TWAI peripheral at 1 Mbit/s. The TJA1051T/3 translates the logic signals to the differential bus [22], [23]. The main cable should form one linear trunk from one physical end to the other. Each motor node connects with a short stub, and CANH/CANL should be routed as a coupled pair with a nearby reference conductor. Exactly two 120 Ω terminations belong at the two physical ends of the assembled bus [25]. Measuring approximately 60 Ω between CANH and CANL with all power removed is a useful static check when both terminators are installed.

The current schematic review identified J9 through J12 as CAN connectors with pin 1 labeled CANH and pin 2 labeled CANL, but the motor harness mapping remained ambiguous and must not be inferred from wire color. The AK40-10 connector and extension harness should be traced end-to-end and compared with the exact actuator manual before power is applied. CAN ground or a reference conductor should accompany the differential pair, while actuator power uses separate conductors sized for motor current.

<INSERT FIGURE 5.4: Draw the physical CAN trunk from the controller/transceiver through all four motor nodes in installed order. Show cable lengths, stub lengths, CANH, CANL, reference/ground, shield if used, node IDs, termination resistor locations, and connector pin numbers>
 What the figure should show: Draw the physical CAN trunk from the controller/transceiver through all four motor nodes in installed order. Show cable lengths, stub lengths, CANH, CANL, reference/ground, shield if used, node IDs, termination resistor locations, and connector pin numbers
 Likely source: final wiring harness drawing and physical measurement

Figure 5.4. Recommended linear CAN topology and termination locations.

5.6 Printed-Circuit-Board Design

The reviewed repository contained KiCad project, schematic, and PCB files under 03_electrical. Earlier PCB evidence showed a controller-header template with no routed segments,

vias, or zones; later schematic work added the power converter and peripheral connectors, but manufacturing and validation were not demonstrated. The final thesis should include the schematic revision, PCB revision, layer stack, copper weight, design-rule check report, electrical-rule check report, fabrication outputs, assembly record, and bring-up results.

Signal routing should preserve an uninterrupted reference plane and keep switch-node copper compact. The LT8609S input capacitor, catch-current loop, inductor, output capacitors, and feedback divider should follow the data-sheet placement guidance [33]. CAN transient protection should sit close to the external connector, and the transceiver decoupling path should be short [23], [25]. IMU placement should minimize vibration and thermal gradients while maintaining a known rigid orientation. High-current actuator paths should be routed as broad copper areas on both outer layers with dense via stitching and without shared narrow necks.

Table 5.4. Recommended printed-circuit-board routing classes

Net class	Track or copper width	Via pad / drill	Clearance	Application and notes
Signals	0.25 mm	0.60 / 0.30 mm	Fabricator default $\geq$ approved minimum	GPIO, UART, SPI, interrupts, CAN logic; route CANH/CANL as a controlled pair where practical.
3.3 V	0.50 mm	0.80 / 0.40 mm	Approved logic clearance	Peripheral distribution; use local planes/pours where helpful.
5 V and LT8609 VIN/VOUT	1.00 mm	1.00 / 0.50 mm	Approved power clearance	Short current loops; verify copper temperature and converter layout.
Each motor branch	5.00 mm top + bottom pour	1.00 / 0.50 mm; 8 vias per net at transitions	0.50 mm minimum; 1.00 mm where possible	One actuator branch; stitch layers approximately every 5 mm.
VBAT and return	10.00 mm top + bottom	1.00 / 0.50 mm; 16 vias per net at transitions	0.50 mm minimum; 1.00 mm where possible	No shared narrow neck; independent input branch for each motor pair.
Each two-motor input branch	pour	0.50 / 0.25 mm; five vias	Per footprint	Confirm against selected package and assembly process.
LT8609 exposed-pad thermal vias	—			

Note. These are design values selected for the present layout, not universal ampacity guarantees. Verify using the final stackup, IPC-2152-based analysis, conductor length, allowed temperature rise, enclosure airflow, and load testing [34]. One-ounce copper is accepted only with two independent XT60 branches, 10 mm top-and-bottom pair pours, 5 mm top-and-bottom motor pours, dense stitching, 14 AWG harness conductors, and no common neck. Two-ounce copper remains preferable.

<INSERT FIGURE 5.5: Insert publication-quality exports of the final schematic sheets. At minimum show power entry and protection, LT8609S regulator, controller headers, IMU, CAN transceiver and termination, four motor connectors, receiver, display, debug interface, test points, and net labels>
 What the figure should show: Insert publication-quality exports of the final schematic sheets. At minimum show power entry and protection, LT8609S regulator, controller headers, IMU, CAN transceiver and termination, four motor connectors, receiver, display, debug interface, test points, and net labels
 Likely source: KiCad schematic plot

Figure 5.5. Final KiCad schematic sheet set.

<INSERT FIGURE 5.6: Insert top and bottom PCB renders with high-current VBAT/return paths, logic planes, CAN pair, switch node, regulator loops, IMU placement, connector orientation, fuses, test points, mounting holes, and thermal vias annotated>
 What the figure should show: Insert top and bottom PCB renders with high-current VBAT/return paths, logic planes, CAN pair, switch node, regulator loops, IMU placement, connector orientation, fuses, test points, mounting holes, and thermal vias annotated
 Likely source: KiCad PCB editor and 3D viewer

Figure 5.6. Final printed-circuit-board layout with power and signal routing identified.

5.7 Electrical Safety and Verification

Initial power-up should use a current-limited bench supply or a fused battery interface with the actuators disconnected. Resistance-to-ground, rail polarity, and regulator output should be checked before installing the controller and modules. Logic rails should then be loaded and observed for startup overshoot and ripple. The CAN transceiver should be verified with the motors unpowered before actuator power is enabled. Each motor branch should be enabled individually while current, connector temperature, and frame error counters are monitored.

The final electrical design should protect against reverse polarity, overcurrent, undervoltage, CAN transients, regulator overheating, and accidental re-arming. The emergency stop must be evaluated as a system function: removing only a software command is not equivalent to isolating actuator energy. The implemented behavior, recovery procedure, and residual powered functions must be documented.

<VERIFY FROM HARDWARE: Record battery manufacturer and part number, cell count, capacity, discharge rating, maximum and minimum operating voltage, connector and wire gauge, fuse types and ratings, master-disconnect hardware, emergency-stop architecture, BMS/protection implementation, converter efficiency and temperatures, rail ripple, peak voltage droop, and measured actuator branch current.>

Chapter 6

KINEMATIC AND DYNAMIC MODELING

6.1 Coordinate Frames and Generalized Coordinates

Let $\{I\}$ denote an inertial frame fixed to the level ground and $\{B\}$ a body frame whose origin lies on the common wheel-center line. The body x axis is forward, y is along the wheel axis, and z is upward at the upright equilibrium. The planar pose is $\eta=[X,Y,\psi]^T$, where ψ is yaw. The body twist is $\nu=[v_x, v_y, \dot{\psi}]^T$. The pitch angle θ describes rotation of the body about the common wheel axis, with $\theta=0$ at the selected upright reference. The generalized coordinates are $q=[X,Y,\psi,\theta]^T$.

$${}^I\mathbf{v} = \mathbf{R}(\psi) {}^B\mathbf{v}, \quad \mathbf{R}(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix} \quad (6.1)$$

Equation (6.1) transforms the body-frame translational velocity to inertial coordinates. The sign of ψ follows the right-hand rule about z_I. All controller and plot signals should use radians internally and state degrees only when presenting results.

<INSERT FIGURE 6.1: Draw the inertial and body frames, pose X-Y-psi, body velocities vx and vy, pitch theta, center of mass, gravity, and common wheel axis. Show the upright reference and positive rotations>
 What the figure should show: Draw the inertial and body frames, pose X-Y-psi, body velocities vx and vy, pitch theta, center of mass, gravity, and common wheel axis. Show the upright reference and positive rotations
 Likely source: MATLAB vector graphic based on final CAD coordinates

Figure 6.1. Inertial and body coordinate frames used by the model.

6.2 Collinear Mecanum-Wheel Kinematics

6.2.1 Wheel contact velocity

Wheel i is located at $p_i=[x_i, y_i]^T$ in $\{B\}$. The planar velocity of its center is the body translation plus the yaw contribution.

$$\mathbf{v}_i^B = \begin{bmatrix} v_x \\ v_y \end{bmatrix} + \dot{\psi} \begin{bmatrix} -y_i \\ x_i \end{bmatrix} \quad (6.2)$$

For the ideal collinear arrangement $x_i=0$. The Mecanum roller constraint is represented by a sign $s_i \in \{-1, +1\}$ for an ideal 45-degree roller. The sign is defined operationally: it specifies whether positive wheel rotation couples positively or negatively to v_y . This

avoids relying on wheel labels such as left-handed or right-handed, whose interpretation can change with viewing direction.

6.2.2 Inverse kinematics

With effective wheel radius r and the stated sign convention, the ideal inverse kinematics reduce to Equation (6.3). Each row states the wheel rate needed for the local longitudinal velocity $v_{x-y_i\dot{\psi}}$ and the lateral component selected by s_i .

$$\boldsymbol{\omega} = \frac{1}{r} \mathbf{H} \boldsymbol{\nu}, \quad \mathbf{H} = \begin{bmatrix} 1 & s_1 & -y_1 \\ 1 & s_2 & -y_2 \\ 1 & s_3 & -y_3 \\ 1 & s_4 & -y_4 \end{bmatrix}, \quad \boldsymbol{\nu} = [v_x \quad v_y \quad \dot{\psi}]^T \quad (6.3)$$

The matrix $\mathbf{H}$ has dimensions 4×3 . The first column corresponds to longitudinal velocity, the second to lateral velocity, and the third to yaw. The equation is symbolic until r , y_i , and s_i are measured. If the roller angle differs materially from 45 degrees, the second column must be scaled from the actual constraint geometry rather than retaining ± 1 .

<INSERT FIGURE 6.2: For each wheel show positive spin, roller axis, effective constrained force direction, sign s_i , axial position y_i , and the local velocity contribution from positive yaw. Use the installed wheel order>
What the figure should show: For each wheel show positive spin, roller axis, effective constrained force direction, sign s_i , axial position y_i , and the local velocity contribution from positive yaw. Use the installed wheel order
Likely source: MATLAB vector diagram checked against the physical wheels

Figure 6.2. Idealized force direction for each collinear Mecanum wheel.

6.2.3 Forward kinematics and rank

When $\mathbf{H}$ has full column rank, the least-squares body twist follows from the Moore-Penrose pseudoinverse.

$$\boldsymbol{\nu} = r (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \boldsymbol{\omega} = r \mathbf{H}^\dagger \boldsymbol{\omega}, \quad \text{rank}(\mathbf{H}) = 3 \quad (6.4)$$

Full planar mobility requires $\text{rank}(\mathbf{H})=3$. Both roller signs must be present; otherwise the longitudinal and lateral columns are dependent. In addition, the axial-position column must not be expressible as a linear combination of the constant and sign columns. For four wheels, a practical condition is that at least one roller-sign group contains wheels at distinct y positions while both signs are represented. These conditions must be evaluated for the measured order rather than assumed.

Table 6.1. Mobility-rank conditions for the collinear wheel arrangement

Geometry or roller case	Expected rank	Consequence
Both signs present and axial positions provide an independent yaw column	3	Independent ideal vx, vy, and yaw commands are kinematically available.
All wheels have the same roller sign	≤ 2	Longitudinal and lateral columns are dependent; full planar mobility is lost.
Each sign occurs only at one repeated axial location	≤ 2	Yaw column can become a combination of longitudinal and lateral columns.
Wheel locations nearly create dependence	3 but ill-conditioned	Large wheel speeds/torques and noise amplification; evaluate condition number.

6.3 Wheel Torque and Body Wrench

Power consistency maps wheel torques to the body wrench $\mathbf{w}=[F_x, F_y, M_z]^T$. If wheel-speed losses are neglected, $\tau^T \omega = \mathbf{w}^T \nu$, giving Equation (6.5).

$$\mathbf{w} = \begin{bmatrix} F_x & F_y & M_z \end{bmatrix}^T = \frac{1}{r} \mathbf{H}^T \boldsymbol{\tau} \quad (6.5)$$

Because four actuators produce three planar wrench components, one internal null-space direction remains when $\mathbf{H}$ has rank three. A weighted minimum-effort allocation is given by Equation (6.6), where $\mathbf{W}$ is positive definite and can penalize actuators differently.

$$\boldsymbol{\tau}^* = r \mathbf{W}^{-1} \mathbf{H} (\mathbf{H}^T \mathbf{W}^{-1} \mathbf{H})^{-1} \mathbf{w}_d \quad (6.6)$$

The allocator must retain torque headroom for pitch stabilization. A planar command that uses all available current can prevent recovery from a pitch disturbance. A practical implementation therefore prioritizes or reserves the longitudinal collective torque needed by the balance loop, then allocates lateral and yaw demands subject to individual current, speed, voltage, and thermal limits. Null-space torque should normally be zero unless it serves a documented objective, because internal wheel forces can increase slip and power loss without changing the ideal body wrench.

6.4 Nonlinear Dynamic Model

The general model uses $\mathbf{q}=[X, Y, \psi, \theta]^T$ and four wheel torques. The Euler-Lagrange form is shown in Equation (6.7). $\mathbf{M}(\mathbf{q})$ is the mass matrix; $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ contains velocity-dependent terms; $\mathbf{g}(\mathbf{q})$ is the gravity vector; $\mathbf{D}$ represents dissipative effects; $\mathbf{B}(\mathbf{q})$ maps actuator torques; and $\mathbf{d}$ contains unmodeled contact, slip, and disturbance forces.

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) + \mathbf{D}\dot{\mathbf{q}} = \mathbf{B}(\mathbf{q})\boldsymbol{\tau} + \mathbf{d} \quad (6.7)$$

A complete numerical $\mathbf{M}$ matrix requires the final component masses, center-of-mass coordinates, and inertia tensors. Until those data exist, the model is kept symbolic. The principal assumptions are a rigid body, a rigid level ground plane, continuous wheel contact,

one dominant unstable pitch coordinate, known wheel geometry, and an idealized roller constraint. The initial model neglects detailed roller inertia, intermittent roller impacts, tire deformation, bearing friction nonlinearities, battery voltage sag, sensor noise, and CAN jitter. Each omitted effect is either included as a disturbance in sensitivity studies or added after hardware data show that it materially affects the result.

Table 6.2. Kinematic and dynamic symbols

Symbol	Definition	Unit
X, Y	Inertial position of body-frame origin	m
ψ	Yaw angle	rad
θ	Pitch angle from upright	rad
v_x, v_y	Body-frame translation velocities	m/s
r	Effective loaded wheel radius	m
y_i	Wheel-i position along common axis	m
s_i	Wheel-i roller sign	—
ω_i	Wheel-i angular speed	rad/s
τ_i	Wheel-i output torque applied in positive wheel direction	N·m
m_b	Body mass participating in pitch motion	kg
m_t	Equivalent translated mass	kg
ℓ	Distance from wheel axis to body COM	m
I_p	Body pitch inertia about COM	kg·m ²
b_x, b_θ	Equivalent longitudinal and pitch damping	N·s/m, N·m·s/rad

6.5 Reduced Longitudinal Balance Model

For initial controller design, longitudinal motion and pitch are isolated from lateral and yaw dynamics. The body center of mass is located a distance ℓ from the wheel axis. The equivalent translated mass m_t includes the body and reflected wheel/rotor contributions used by the chosen derivation. The collective wheel torque is $\tau_\Sigma = \sum \tau_i$ for the wheel combination that produces positive forward motion. With positive θ defined as forward lean and positive wheel torque producing forward ground force, one consistent reduced model is Equation (6.8).

$$\begin{bmatrix} m_t & m_b \ell \cos \theta \\ m_b \ell \cos \theta & I_p + m_b \ell^2 \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} -m_b \ell \sin \theta \dot{\theta}^2 \\ -m_b g \ell \sin \theta \end{bmatrix} + \begin{bmatrix} b_x \dot{x} \\ b_\theta \dot{\theta} \end{bmatrix} = \begin{bmatrix} 1/r \\ -1 \end{bmatrix} \tau_\Sigma \quad (6.8)$$

The input vector contains the forward force τ_Σ/r and the opposing reaction torque $-\tau_\Sigma$ on the body. The equation assumes that the four wheel torques contributing to forward balance have already been transformed into a consistent sign convention. If the actuator reports motor-side rather than output torque, gear ratio and efficiency must be applied explicitly. The model should be checked by energy and dimensional consistency before numerical use.

<INSERT FIGURE 6.3: Draw the reduced body and wheel system with body mass m_b , COM distance l , pitch inertia I_p , wheel radius r , gravity, ground force, collective actuator torque, body reaction torque, x displacement, and positive θ >
What the figure should show: Draw the reduced body and wheel system with body mass m_b , COM distance l , pitch inertia I_p , wheel radius r , gravity, ground force, collective actuator torque, body reaction torque, x displacement, and positive θ
Likely source: author-created free-body diagram verified against the sign convention

Figure 6.3. Reduced inverted-pendulum free-body representation.

6.6 Linearization and State-Space Form

Near $\theta=0$, $\sin\theta\approx\theta$ and $\cos\theta\approx 1$. Let E be the 2×2 mass matrix evaluated at the upright equilibrium, K contain the linear gravitational term, and D contain the selected viscous damping. Defining $\mathbf{x}=[x, \theta, \dot{x}, \dot{\theta}]^T$ gives the continuous-time model in Equation (6.9).

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u, \quad \mathbf{x} = [x \quad \theta \quad \dot{x} \quad \dot{\theta}]^T, \quad \mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{E}^{-1}\mathbf{K} & -\mathbf{E}^{-1}\mathbf{D} \end{bmatrix} \quad (6.9)$$

The exact A and B entries should be generated symbolically from the adopted parameter definitions to avoid transcription errors. The model is discretized at the measured controller interval, preferably using a zero-order hold. Controllability must be checked with the final numerical parameters using Equation (6.10); observability must likewise be checked for the selected measured outputs.

$$\mathcal{C} = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B} \quad \mathbf{A}^3\mathbf{B}], \quad \text{rank}(\mathcal{C}) = 4 \text{ required} \quad (6.10)$$

Table 6.3. Model parameters and identification method

Parameter	Source for final value	Uncertainty or validation method
m_b and m_t	Physical component weights and final CAD grouping	Scale accuracy; compare CAD and assembled mass
ℓ and COM offsets	CAD mass properties and physical balance/suspension test	Repeat after wiring or battery changes
I_p and I_z	CAD plus pendulum/system-identification test	Report method and confidence interval
r	Loaded rollout test	Repeat on test surface and at representative load
y_i and s_i	Dimensioned CAD and wheel-direction inspection	Cross-check against single-wheel command test
Actuator delay/time constant	Command/feedback step test	Fit model; report sample rate and confidence
Torque/current limits	Installed actuator configuration and measured current	Distinguish manufacturer rating from selected limit
b_x and b_θ	Coast-down or fitted response	Treat as uncertain in sensitivity analysis
Friction/slip parameters	Dedicated velocity/force experiments	Surface-dependent; do not overfit one trial

6.7 Model Limitations

The ideal model predicts achievable velocity and generalized wrench, not exact ground forces at every roller. Mecanum wheels experience discrete changes in the contacting roller, varying normal loads, and slip [10]–[11]. The body may also roll or flex even when the principal unstable motion is pitch. The actuator's internal current and velocity loops, CAN command update, gearbox backlash, and encoder location modify the commanded-to-wheel response. These effects should be added only at a level supported by measurement.

The purpose of the reduced model is to make controller assumptions visible and testable. Agreement should be evaluated over multiple initial conditions and command directions. If one parameter set cannot explain both small-signal balance and larger motion trials, the residuals should guide model refinement rather than being hidden by retuning the controller.

Chapter 7

CONTROL AND STATE ESTIMATION

7.1 Current Implementation Status

The inspected firmware contains types and state paths needed by a balance controller, including pitch, pitch rate, roll, and roll rate, but no completed balance-control module was confirmed. The current system should therefore be described as control-ready infrastructure rather than a demonstrated self-balancing implementation. The architecture in this chapter is the proposed completion path and the basis of the simulation draft. Final controller claims require the exact source revision, gains, sample interval, actuator mode, and hardware logs.

Table 7.1. Control and estimation functions and current status

Function	Current evidence	Status used in thesis	Completion evidence
Raw IMU acquisition	Portable ISM330DHCX driver and ESP32 SPI path	Implemented/bring-up	Rate, missed-sample, and noise measurements
Mount transform and gyro calibration	Reported in firmware data path	Implemented, final calibration unverified	Axis test and retained calibration values
Complementary attitude estimate	Reported estimator path	Implemented infrastructure	Static/dynamic comparison with reference angle
Balance-state snapshot	Balance types and application state reported	Implemented interface	Atomicity and freshness test
Balance control law	No confirmed controller source	Planned	Code, gains, timing, simulation and hardware results
Wheel wrench allocation	No confirmed mapping/roller signs	Planned	Geometry matrix, sign test, saturation logic
Yaw/lateral outer loops	No confirmed closed-loop implementation	Planned	Code and motion-tracking results
Safety-gated motor output	Supervisor/safety command gating reported	Partially implemented	Fault-injection and stop-time results

7.2 Proposed Cascaded Architecture

The proposed controller separates stabilization, planar motion, allocation, and actuator enforcement. The innermost balance loop uses pitch and pitch rate, with translational velocity or position included directly in a state-feedback law or applied through a slower outer loop. Lateral and yaw controllers produce desired body forces or moments. An allocation stage maps the combined body command to four wheel commands, enforces per-actuator limits, and reserves balance authority. A supervisor controls transitions among disarmed, ready, balancing, and faulted states.

<INSERT FIGURE 7.1: Draw reference conditioning, x/y/yaw outer loops, pitch estimator, inner balance controller, body-wrench command, four-wheel allocator, saturation and slew limits, safety permission, CAN actuator interface, and feedback paths. Mark implemented versus planned blocks>
What the figure should show: Draw reference conditioning, x/y/yaw outer loops, pitch estimator, inner balance controller, body-wrench command, four-wheel allocator, saturation and slew limits, safety permission, CAN actuator interface, and feedback paths. Mark implemented versus planned blocks
Likely source: firmware architecture and MATLAB controller model

Figure 7.1. Proposed cascaded control and allocation architecture.

The controller should not command position while disarmed or fallen. On entry to balancing, integrators and reference states should be initialized from the measured state to prevent a command step. Operator inputs should be deadbanded, scaled, rate limited, and set to zero when CRSF data become stale. Display and telemetry functions may observe control state but may not own motor commands.

7.3 State Estimation

7.3.1 Calibration and frame transformation

The IMU driver first converts raw counts to physical units using the configured full-scale ranges. A fixed mounting matrix transforms sensor axes into the body frame. Gyro bias is estimated only while the robot is confirmed stationary, and the calibration record includes sample count, duration, temperature, mean, and standard deviation. Accelerometer offsets and scale can be checked with multiple static orientations. The estimator marks its output invalid during initialization, sensor errors, or stale acquisition.

7.3.2 Complementary pitch estimate

Let ω_p be the body-frame gyro component about the pitch axis, θ_a the gravity-derived pitch angle, and Δt the measured sample interval. A discrete complementary filter is given by Equation (7.1).

$$\hat{\theta}_k = \alpha (\hat{\theta}_{k-1} + \Delta t \omega_{p,k}) + (1 - \alpha) \theta_{a,k} \quad (7.1)$$

The coefficient α determines the crossover between the integrated gyro and accelerometer. It should be computed from a documented time constant and actual Δt , not copied as a unitless tuning constant without context. During forward or lateral acceleration, the accelerometer does not measure gravity alone; the accelerometer correction can therefore tilt the estimate in the direction of specific force. Validation should compare the estimated pitch with an external reference during static tilts, manual rotations, motor acceleration, and disturbance recovery.

<INSERT FIGURE 7.2: Show raw accelerometer and gyroscope samples, unit conversion, mounting transform, bias calibration, gravity angle, gyro integration, complementary fusion, validity/freshness checks, and the published pitch/pitch-rate snapshot>

What the figure should show: Show raw accelerometer and gyroscope samples, unit conversion, mounting transform, bias calibration, gravity angle, gyro integration, complementary fusion, validity/freshness checks, and the published pitch/pitch-rate snapshot

Likely source: firmware data-flow diagram and estimator implementation

Figure 7.2. Attitude-estimation signal path from raw IMU data to balance state.

7.4 Balance Control

7.4.1 State-feedback option

For the discrete linear model, the initial proposed law is Equation (7.2), where $\hat{\mathbf{x}}$ contains estimated longitudinal and pitch states, $\mathbf{r}$ contains motion references, $\mathbf{K}$ is the state-feedback gain, and $\mathbf{N}_r$ supplies reference feedforward or integral augmentation.

$$\mathbf{u}_k = -\mathbf{K}\hat{\mathbf{x}}_k + \mathbf{N}_r\mathbf{r}_k, \quad \mathbf{K} = \text{dlqr}(\mathbf{A}_d, \mathbf{B}_d, \mathbf{Q}, \mathbf{R}) \quad (7.2)$$

The continuous model is discretized at the measured control interval. $\mathbf{Q}$ and $\mathbf{R}$ are selected after scaling the states to meaningful limits so that numerical magnitude does not substitute for engineering priority. The eigenvalues of the closed-loop discrete system, control effort, saturation time, and robustness to parameter changes are reported. Integral action may be added for steady velocity or position error only after the inner balance loop is stable and anti-windup is implemented.

7.4.2 PID option and anti-windup

A cascaded PID implementation remains a valid alternative for initial hardware tuning. The pitch controller uses angle error, measured or filtered pitch rate, and carefully limited integral action. An outer velocity loop modifies the pitch reference. Equation (7.3) shows a continuous representation with back-calculation anti-windup.

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \dot{e}(t), \quad \dot{z} = e + K_{aw} (u_{sat} - u) \quad (7.3)$$

Derivative action should use measured angular rate or a filtered derivative rather than differentiating a noisy angle. The integral state is reset or frozen when disarmed, fallen, or saturated in a direction that would increase windup. The final thesis should report which controller was implemented and use the other only as a comparison if it was actually tested.

7.5 Lateral, Yaw, and Motion Control

Longitudinal force is coupled most directly to pitch. Lateral and yaw commands are generated in body coordinates and passed to the allocator with lower priority than the balance channel. The lateral controller may begin as open-loop velocity command scaling because roller slip can make encoder-only lateral speed estimation unreliable. Closed-loop lateral velocity requires an independent velocity estimate or a validated wheel-slip model. Yaw can be estimated from the gyro and corrected with an external heading reference if required by the experiment.

The allocator uses Equation (6.6) with measured H . Each wheel command is converted to the actuator's selected operating mode. If the AK40-10 is operated in servo velocity mode, the balance design must account for the actuator's internal velocity loop and command period. If torque or MIT mode is selected, torque scaling, stiffness, damping, and current limits must be documented. Mode selection is not finalized in the available evidence.

7.6 Saturation, Rate Limits, and Safety

Command limits are applied in physical units after allocation. The implemented order should be documented because independent clipping can distort the desired wrench. Equation (7.4) represents a rate limit followed by magnitude saturation; the actual design may use a constrained allocator so that all limits are considered together.

$$\tau_{cmd} = \text{sat}_{\tau_{max}}(\text{rateLimit}(\tau^*, \dot{\tau}_{max})) \quad (7.4)$$

Safety checks include IMU validity, estimate age, pitch and pitch-rate bounds, actuator-feedback age, CAN state, battery voltage, motor faults, operator link state, emergency-stop state, and explicit arming intent. A fault removes torque permission and requests a defined safe motor state. Automatic re-arming after a fault is prohibited unless the final hazard analysis supports it.

<INSERT FIGURE 7.3: Show DISARMED, INITIALIZING, READY, BALANCING, FALLEN, and FAULT states with guarded transitions for calibration completion, arm request, valid sensors, pitch window, stale data, motor fault, emergency stop, and manual reset>

What the figure should show: Show DISARMED, INITIALIZING, READY, BALANCING, FALLEN, and FAULT states with guarded transitions for calibration completion, arm request, valid sensors, pitch window, stale data, motor fault, emergency stop, and manual reset

Likely source: supervisor state-machine implementation

Figure 7.3. Safety-oriented operating-state transitions.

Table 7.2. Controller and estimator parameters requiring final values

Parameter	Unit	Final value	Selection/verification method
Estimator sample period	s	<INSERT>	Measured timestamps and configured IMU ODR
Control sample period	s	<INSERT>	Schedulability and timing measurement
Complementary time constant / α	s / —	<INSERT>	Noise and external-angle comparison
LQR Q and R or PID gains	various	<INSERT>	Document scaling, simulation, restrained tuning, final code revision
Torque/current limit per actuator	N·m / A	<INSERT>	Installed motor configuration and thermal/current tests
Command slew limit	N·m/s or mode-specific	<INSERT>	Avoid current spikes while preserving recovery
Arm pitch window	rad	<INSERT>	Safe test procedure and recovery capability
Fall pitch threshold	rad	<INSERT>	Mechanical clearance and loss-of-recovery evidence
Sensor/feedback timeout	s	<INSERT>	Worst-case timing and fault-injection test

7.7 Stability and Validation

For a linear controller, local stability is assessed from the closed-loop poles or eigenvalues of the discrete system. This analysis is necessary but insufficient because saturation, rate limiting, delay, estimator error, and slip are nonlinear. Simulation should sweep mass, center-of-mass height, inertia, actuator delay, friction, and battery-dependent limits. Hardware validation then begins with low torque and a restraint, advances to brief free-standing trials, and records every fall or fault.

<INSERT RESULT: Final controller type, equations as implemented, exact state vector, sample interval, gains, filters, reference limits, saturation logic, stability analysis, simulation response, and hardware response. Cite the Git commit and configuration file containing each parameter.>

Chapter 8

EMBEDDED FIRMWARE AND SOFTWARE

8.1 Repository Organization

The repository separates portable common logic from platform-specific implementations under `04_firmware` [9]. The later architecture includes `common/`, `platforms/esp32/`, and `platforms/stm32/`. The common layer contains interfaces, device-independent drivers, estimation and control types, and application state; the platform layer owns MCU peripherals, RTOS tasks, pin configuration, and build files. This structure supports the stated goal of bringing up the robot on an ESP32 while preserving a path to the STM32H723ZG.

Earlier STM32 development material also contains CubeMX-generated Core, Drivers, and Middlewares trees, CMake and linker support, and a `zephyr.ioc` configuration. Those files represent a historical or scaffolded platform and should not be described as the active robot firmware unless the final repository establishes that status. The final thesis should name the exact target and commit used for every experiment.

<INSERT FIGURE 8.1: Create a directory tree for the final tested commit. Distinguish portable common code, ESP32 BSP and tasks, STM32 scaffold or implementation, configuration headers, tests/bring-up code, documentation, and generated vendor code. Mark historical directories>
What the figure should show: Create a directory tree for the final tested commit. Distinguish portable common code, ESP32 BSP and tasks, STM32 scaffold or implementation, configuration headers, tests/bring-up code, documentation, and generated vendor code. Mark historical directories
Likely source: repository tree at the final experiment commit

Figure 8.1. Firmware directory and portability layers.

8.2 Sensor Acquisition and Estimation Path

The portable IMU interface uses `common/include/interfaces/if_spi.h`, `common/include/drivers/drv_ism330dhcx.h`, and `common/src/drivers/drv_ism330dhcx.c` in the previously inspected structure. The driver reads the ISM330DHCX identity register and configures accelerometer and gyroscope control registers. A reported configuration used 208 Hz output data rate, ± 4 g acceleration range, ± 500 degrees/s angular-rate range, and a 14-byte burst beginning at the temperature/gyro/accelerometer output block. These values must be rechecked against the final source and the ISM330DHCX data sheet [30].

The higher-level path is SPI BSP $\rightarrow$ portable IMU driver $\rightarrow$ mounting transform $\rightarrow$ gyro calibration $\rightarrow$ attitude estimator $\rightarrow$ timestamped IMU snapshot $\rightarrow$ safety check $\rightarrow$ balance-state extraction. The snapshot should include a monotonically increasing timestamp, validity flag, sequence counter or update marker, and diagnostic status. A consumer should copy one consistent snapshot rather than read fields that can change during the calculation.

8.3 Radio, Display, and User Interaction

The RadioMaster RP2 V2 receiver uses the CRSF serial protocol. The current firmware reportedly decodes 16 channels and monitors link state. CRSF framing and channel definitions are available in the ExpressLRS source [35], while the installed receiver manual and configuration should be archived with the project [31]. The production command path should map only documented channels, apply range checks and deadbands, and force a neutral command when a valid frame is not received within the selected timeout.

The ST7789 display provides local bring-up and operating status [32]. UI work should run at lower priority than sensing, estimation, control, and motor communication. It should consume published state rather than directly reading hardware owned by another task. Diagnostic pages should show mode, arm state, pitch, IMU status, radio status, battery state, four actuator states, CAN faults, and timing without allowing the display driver to block a critical loop.

8.4 CAN and Actuator Path

The ESP32 platform uses TWAI at 1 Mbit/s and communicates with CubeMars actuators. The firmware includes a four-slot actuator snapshot and feedback/fault monitoring. The inspected configuration identified motor 1 with CAN ID 69 while the remaining IDs or enable flags were not confirmed. Servo-velocity commands were issued on a reported 20 ms period in `config_actuator.h`. That interval is evidence of the current bring-up configuration, not necessarily the final balance-loop rate.

The motor path should have one command owner. The controller publishes desired physical commands, safety publishes torque permission and fault state, and the motor task transmits the approved result. Bring-up tests that directly transmit actuator frames should be compiled or enabled only in an explicit test build. Feedback parsing should reject invalid lengths and identifiers, apply mode-specific scaling, timestamp every update, and expose motor faults without clearing them silently.

<INSERT FIGURE 8.2: Show producers and consumers of IMU samples, estimated state, CRSF command, supervisor mode, safety permission, desired wrench, allocated wheel commands, CAN frames, actuator feedback, UI state, and logging. Identify snapshot, queue, notification, or direct-call boundaries>
What the figure should show: Show producers and consumers of IMU samples, estimated state, CRSF command, supervisor mode, safety permission, desired wrench, allocated wheel commands, CAN frames, actuator feedback, UI state, and logging. Identify snapshot, queue, notification, or direct-call boundaries
Likely source: final firmware architecture and source review

Figure 8.2. Runtime data flow through sensing, estimation, supervision, control, and actuation.

8.5 Initialization and Runtime Scheduling

A safe initialization order begins with outputs in inactive states, configuration loading, time-base and logging initialization, communication peripherals, sensor identity checks, actuator bus startup, calibration, and task creation. The supervisor remains disarmed until the IMU estimate is valid, actuator feedback is fresh, the operator link is valid, no safety fault is active, and pitch is within the arm window. Startup should never preserve a prior nonzero command.

<INSERT FIGURE 8.3: Draw the exact initialization sequence from reset to DISARMED and READY. Include GPIO safe levels, SPI/IMU identity, gyro calibration, TWAI start, motor discovery, radio start, display start, configuration checks, snapshot validity, and arm conditions>

What the figure should show: Draw the exact initialization sequence from reset to DISARMED and READY. Include GPIO safe levels, SPI/IMU identity, gyro calibration, TWAI start, motor discovery, radio start, display start, configuration checks, snapshot validity, and arm conditions

Likely source: final main entry point, task creation, and supervisor code

Figure 8.3. Firmware initialization and arming sequence.

Table 8.1. Runtime tasks and timing requirements

Function/task	Trigger or period	Priority/deadline	Worst-case time	Status
IMU acquisition and estimation	<VERIFY; IMU configured ODR and task/event trigger>	Highest sensing/control class	<MEASURE>	Implemented
Balance control	<SELECT AND MEASURE>	Highest control class	<MEASURE>	Planned
Motor command/feedback	Reported 20 ms servo command period; final pending	High	<MEASURE>	Partial
CRSF receive/decode	UART/event-driven or configured poll	Medium-high	<MEASURE>	Implemented
Supervisor/safety	<VERIFY>	High enough to enforce timeout	<MEASURE>	Infrastructure reported
Display/UI	<VERIFY; target 10–20 Hz if retained>	Low	<MEASURE>	Implemented
Telemetry/logging	Buffered/event-driven	Low	<MEASURE including backpressure>	Incomplete or unverified

Note. Earlier STM32 prototypes used task periods such as 2 ms CRSF, 10 ms IMU, 50 ms OLED, and 200 ms application UI. Those historical values are not attributed to the current ESP32 build without source confirmation.

<INSERT FIGURE 8.4: Create a measured timing diagram over several control cycles showing IMU data-ready or read, estimator completion, control computation, allocation, CAN transmit, actuator feedback, radio processing, and low-priority UI/logging. Include minimum, mean, maximum, and jitter>
 What the figure should show: Create a measured timing diagram over several control cycles showing IMU data-ready or read, estimator completion, control computation, allocation, CAN transmit, actuator feedback, radio processing, and low-priority UI/logging. Include minimum, mean, maximum, and jitter
 Likely source: firmware timestamps, logic analyzer, and CAN trace

Figure 8.4. Control-cycle timing and communication schedule.

8.6 Safety, Fault Handling, and Data Freshness

Safety is implemented as explicit state and permission, not as scattered limit checks. Each data product carries a timestamp and validity. The safety logic compares age against a source-specific limit, checks numeric range and finiteness, and latches faults that require operator inspection. Motor feedback faults, bus-off, IMU failure, radio loss, low battery, excessive pitch, and emergency stop all remove actuator permission.

Table 8.2. Proposed operating and safety states

State	Entry condition	Permitted actuator behavior	Exit condition
DISARMED	Reset, user disarm, or safe recovery	Torque disabled or documented zero-output mode	Valid system and explicit prepare/arm sequence
INITIALIZING	Peripheral and estimator startup	Disabled	Calibration and device checks complete or fault
READY	Valid sensors and motors; within arm window	Disabled	Explicit arm request or new fault
BALANCING	Arm request with all guards true	Safety-approved controller command	Disarm, fall, stale data, or fault
FALLEN	Pitch/rate exceeds recovery boundary	Disabled/zero	Manual reset after safe repositioning
FAULT	Electrical, communication, sensor, or software fault	Disabled/zero	Fault cleared and explicit manual reset

<INSERT FIGURE 8.5: Trace one desired motor command through mode checks, data-age checks, pitch limits, E-stop, actuator limits, packing, TWAI transmission, feedback timeout, and fault transition. Identify the single function allowed to submit a production CAN command>

What the figure should show: Trace one desired motor command through mode checks, data-age checks, pitch limits, E-stop, actuator limits, packing, TWAI transmission, feedback timeout, and fault transition. Identify the single function allowed to submit a production CAN command

Likely source: final motor task and safety code

Figure 8.5. Motor-command safety gate and fault response.

8.7 Configuration, Testing, and Version Control

Hardware-dependent values should be centralized in configuration headers or generated platform configuration: pin assignments, CAN bitrate and identifiers, IMU ranges and sample rate, receiver UART settings, display pins, actuator mode, command period, limits, and safety timeouts. Values should use physical units at module boundaries. Compile-time assertions and startup logs can detect invalid combinations.

Production and bring-up paths should remain separate. A bring-up build may contain direct tests for debug UART, CRSF, IMU identity, display pages, CAN loopback, and one-motor motion. The normal build should start through the supervisor and safety path. Each thesis experiment should record the Git commit, build target, configuration checksum or file revision, actuator firmware/settings, and any local changes.

Table 8.3. Firmware modules and repository evidence

Repository path or module	Responsibility	Evidence at inspected snapshot	Thesis status
04_firmware/common/include/interfaces/if_spi.h	Portable SPI abstraction	Previously inspected	Implemented structure
04_firmware/common/include/drivers/drv_ism330dhcx.h	IMU public interface	Previously inspected	Implemented
04_firmware/common/src/drivers/drv_ism330dhcx.c	ISM330DHCX register and conversion logic	Previously inspected	Implemented/verify final configuration
04_firmware/common/include/control/ctrl_balance_types.h	Balance-state/control data types	Observed at commit 97e5a0a	Interface only; controller absent
04_firmware/platforms/esp32	Active BSP, tasks, and peripheral integration	Reported working bring-up platform	Partial system
04_firmware/platforms/esp32/main/config/config_actuator.h	Actuator configuration and command period	20 ms servo-velocity period previously observed	Bring-up setting; final value pending
04_firmware/platforms/stm32 and/or generated STM32 tree	Future STM32 implementation	Scaffold/generated code	Planned migration
03_electrical/Zephyr.kicad_*	Controller PCB design	Schematic/PCB project present	Design in progress
05_simulation	Current repository simulation directory	Visible at repository root; contents not fully re-inspected	Requires ZIP/current connector verification

<VERIFY FROM REPOSITORY: Replace the summarized paths in Table 8.1 with the exact final tree, task names, line-referenced configuration values, and tested commit. The current GitHub directory traversal was not available to the document-generation environment even after connection; upload a ZIP containing .git metadata or git log output if branch and commit-history evidence is required.>

Chapter 9

SIMULATION AND ANALYSIS

9.1 Objectives and Status

Simulation serves four purposes: verify the collinear kinematic signs and rank; test the nonlinear balance equations; select an initial controller without risking hardware; and identify which parameters most strongly affect performance. A MATLAB script named `simulate_collinear_mecanum_robot.m` was developed in prior project work as a draft artifact. It used four mechanical coordinates $q=[X,Y,\psi,\theta]$, eight mechanical states, four actuator-lag states, a nonlinear Euler-Lagrange model, an LQR controller, sample-and-hold control, fourth-order Runge-Kutta integration, saturation, slew limits, and DISARMED/BALANCING/FALLEN logic. That script was created in a prior sandbox and was not confirmed in the repository snapshot; any numerical examples from it are not thesis results.

<INSERT FIGURE 9.1: Show parameter loading, geometry/Jacobian construction, rank checks, nonlinear plant, actuator lag and saturation, sampled controller, state machine, RK4 integration, logging, metrics, plots, and animation>

What the figure should show: Show parameter loading, geometry/Jacobian construction, rank checks, nonlinear plant, actuator lag and saturation, sampled controller, state machine, RK4 integration, logging, metrics, plots, and animation
Likely source: final MATLAB simulation code

Figure 9.1. Simulation model and numerical-integration workflow.

9.2 Model Structure and Numerical Integration

The simulation state is $\mathbf{x}_s=[X,Y,\psi,\theta,\dot{X},\dot{Y},\dot{\psi},\dot{\theta},\tau_1,\dots,\tau_4]^T$ when first-order actuator torque states are retained. Parameters are defined at the beginning of the script with units and source fields. The controller executes at its discrete sample period while the nonlinear plant is integrated at a smaller step. A zero-order hold preserves the last command between controller updates.

Fourth-order Runge-Kutta integration is represented by Equation (9.1). Step-size convergence is checked by repeating each baseline case at half the integration step and comparing key metrics. Solver success alone is not evidence of model correctness; dimensional checks, energy checks for the unforced plant, zero-input equilibrium, forward/inverse kinematic round trips, and sign tests are performed before interpreting a response.

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \frac{\Delta t}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \quad (9.1)$$

<INSERT FIGURE 9.2: Render the robot body, four collinear wheel centers, individual wheel rotation, roller handedness, inertial and body axes, center of mass, pitch lean, yaw, and ground trajectory. Display time and operating state without implying unmeasured physical geometry>
What the figure should show: Render the robot body, four collinear wheel centers, individual wheel rotation, roller handedness, inertial and body axes, center of mass, pitch lean, yaw, and ground trajectory. Display time and operating state without implying unmeasured physical geometry
Likely source: MATLAB visualizer using measured CAD geometry

Figure 9.2. Three-dimensional visualization of the collinear Mecanum robot model.

9.3 Kinematic Validation

Kinematic tests begin with unit commands in v_x , v_y , and $\dot{\psi}$. Equation (6.3) generates wheel rates, and Equation (6.4) reconstructs the body twist. The residual norm, H condition number, and sign pattern are reported. Combined commands then test superposition. These tests validate the algebra only; they do not demonstrate traction or wheel-ground behavior.

Table 9.1. Simulation cases and required outputs

Case	Initial condition/input	Signals and plot	Derived metric
K1 longitudinal	Unit $+v_x$; $v_y=\dot{\psi}=0$	Four wheel rates; reconstructed twist	Round-trip error and signs
K2 lateral	Unit $+v_y$; $v_x=\dot{\psi}=0$	Four wheel rates; reconstructed twist	Round-trip error and roller pattern
K3 yaw	Unit $+\dot{\psi}$; $v_x=v_y=0$	Wheel rates versus y_i	Round-trip error and yaw leverage
K4 combined	Bounded multiaxis command	Requested/reconstructed twist	Maximum and RMS error
D1 small pitch	Small initial θ ; zero motion reference	θ , $\dot{\theta}$, x , command, state	Stability, settling, saturation
D2 disturbance	Defined impulse or initial angle	Pitch and command	Recoverable disturbance/angle
D3 trajectory	Bounded x-y-yaw reference	Pose, velocities, pitch, wheel commands	Tracking error and torque reserve
S1 sensitivity	Parameter and delay sweeps	Metric versus parameter	Robustness margins/trends

<INSERT FIGURE 9.3: Create three panels for pure longitudinal, pure lateral, and pure yaw commands. Horizontal axis: time in seconds. Vertical axes: commanded and reconstructed body velocities in m/s and yaw rate in rad/s; include four wheel speeds in rad/s. Use the controller sample interval and at least ten samples per fastest displayed transient>
What the figure should show: Create three panels for pure longitudinal, pure lateral, and pure yaw commands. Horizontal axis: time in seconds. Vertical axes: commanded and reconstructed body velocities in m/s and yaw rate in rad/s; include four wheel speeds in rad/s. Use the controller sample interval and at least ten samples per fastest displayed transient
Likely source: MATLAB kinematic validation logs

Figure 9.3. Kinematic validation for independent longitudinal, lateral, and yaw commands.

9.4 Dynamic and Controller Cases

The first dynamic case starts within the linear region with no planar-motion request. The second applies a documented initial pitch or torque disturbance. The third adds low-amplitude longitudinal commands, followed by lateral and yaw commands after balance remains stable. Saturation, state transitions, and actuator states appear on every plot so that apparent tracking is not mistaken for unconstrained performance.

<INSERT FIGURE 9.4: Plot time in seconds on the horizontal axis. Use vertically aligned panels for pitch angle in degrees, pitch rate in degrees/s, longitudinal position/velocity, collective command in N·m or mode-specific units, individual wheel commands, saturation flag, and operating state. Compare nonlinear and linear predictions if both are simulated>
What the figure should show: Plot time in seconds on the horizontal axis. Use vertically aligned panels for pitch angle in degrees, pitch rate in degrees/s, longitudinal position/velocity, collective command in N·m or mode-specific units, individual wheel commands, saturation flag, and operating state. Compare nonlinear and linear predictions if both are simulated
Likely source: MATLAB simulation output from the final parameter set

Figure 9.4. Simulated upright-balance response for the selected controller.

9.5 Sensitivity and Uncertainty

Sensitivity cases vary assembled mass, center-of-mass height, pitch inertia, effective radius, actuator delay, torque limit, feedback age, damping, and roller-force effectiveness over documented ranges. One-factor sweeps show local trends; selected combinations expose interactions. The ranges should come from measurement uncertainty or credible configuration bounds, not arbitrary percentages.

<INSERT FIGURE 9.5: Use an x-axis of the varied parameter with units and y-axes for RMS pitch error, peak pitch, settling time, maximum torque/current, and stability/fall outcome. Include the nominal case and uncertainty bounds; compare delay, COM height, inertia, torque limit, and lateral-force effectiveness>
What the figure should show: Use an x-axis of the varied parameter with units and y-axes for RMS pitch error, peak pitch, settling time, maximum torque/current, and stability/fall outcome. Include the nominal case and uncertainty bounds; compare delay, COM height, inertia, torque limit, and lateral-force effectiveness
Likely source: MATLAB parameter-sweep results

Figure 9.5. Sensitivity of balance response to mass properties, delay, and actuator limits.

Table 9.2. Simulation parameter record

Parameter	Value	Unit	Source/revision	Status
Wheel radius	<INSERT>	m	Loaded rollout/CAD	Required
Wheel coordinates and signs	<INSERT>	m / —	Final assembly	Required
Body and total mass	<INSERT>	kg	Scale/CAD	Required
COM height and offsets	<INSERT>	m	CAD/physical test	Required
Pitch and yaw inertias	<INSERT>	kg·m ²	CAD/identification	Required
Actuator response and delay	<INSERT>	s	Bench step test	Required
Torque/current/speed limits	<INSERT>	N·m / A / rad/s	Installed settings	Required
Controller period	<INSERT>	s	Firmware measurement	Required
Integrator step	<INSERT>	s	Convergence study	Required
Initial conditions	<INSERT>	SI units	Case definition	Required

9.6 Model-to-Hardware Comparison

After hardware tests, simulation and measured data should share the same reference commands, initial conditions, sampling grid, sign convention, and units. Compare pitch, pitch rate, wheel or actuator feedback, body velocity, and command. Quantify time alignment and filtering. Fit only parameters designated for identification, then validate the fitted model on different trials. Reporting only a visually similar trace would not establish predictive validity.

<INSERT DATA: Provide final MATLAB code, parameter file, raw simulation logs, software release, solver settings, sample interval, integration step, and reproducibility instructions. Replace all example values with measured or cited parameters.>

Chapter 10

PROTOTYPE INTEGRATION AND EXPERIMENTAL METHODS

This chapter separates completed subsystem bring-up from the experimental work still required to validate an integrated, freely balancing robot. Repository evidence and project records support operation of the four actuators, IMU acquisition and estimation, CRSF receiver decoding, display output, CAN/TWAI communication, and supervisor infrastructure. They do not yet support a claim of closed-loop balancing or omnidirectional motion. Consequently, Sections 10.1 and 10.2 document the present integration baseline, while the remaining sections specify proposed procedures in future tense.

<INSERT FIGURE 10.1: Photograph the assembled robot on a neutral background. Label the four collinear Mecanum wheels, actuator modules, frame, battery, controller, IMU, CAN harness, receiver, display, center-of-mass region, balancing axis, and body axes. Include a metric scale and image date>
What the figure should show: Photograph the assembled robot on a neutral background. Label the four collinear Mecanum wheels, actuator modules, frame, battery, controller, IMU, CAN harness, receiver, display, center-of-mass region, balancing axis, and body axes. Include a metric scale and image date
Likely source: dated photograph of the final physical prototype

Figure 10.1. Integrated prototype prepared for controlled bench testing.

10.1 Integration Baseline

The confirmed development platform uses an ESP32-class controller and FreeRTOS-based firmware. The four selected AK40-10 actuators can be commanded during bring-up, and the firmware contains four actuator slots; however, the inspected configuration explicitly identified only one active node address. The ISM330DHCX driver reads a contiguous sample containing accelerometer and gyroscope data, applies a mounting transform, calibrates gyro bias, and supplies a complementary attitude estimator. Receiver and display paths have also been exercised. These achievements establish interface readiness but not timing determinism, calibrated dynamics, or closed-loop stability.

Table 10.1. Integration and experimental test matrix

ID	Test	Present status	Acceptance evidence	Disposition
I-01	Mechanical assembly and wheel-order survey	Partially documented	Dimensioned drawing, photographs, torque record	Complete before modeling freeze
I-02	Power-off continuity and insulation	No record located	Signed inspection sheet	Pending
I-03	Current-limited regulator bring-up	No final-PCB record located	Rail voltage/ripple plots	Pending
I-04	CAN physical layer and four-node enumeration	Single-node evidence; four slots in code	Oscilloscope trace and node table	Pending
I-05	IMU identity, axes, bias, and time stamps	Driver/estimator implemented	Calibration record and sign tests	Repeat on final mount
I-06	Receiver failsafe and display	Subsystem operation reported	Recorded timeout and UI test	Document
I-07	Restrained balance loop	Not confirmed	Synchronized closed-loop log	Pending
I-08	Free balance and disturbance rejection	Not confirmed	Repeated trials and metrics	Pending
I-09	Longitudinal/lateral/yaw motion	Not confirmed	Reference and measured trajectories	Pending
I-10	Power, thermal, endurance, and fault tests	Not confirmed	Logged voltage/current/temperature/errors	Pending

Note. A completed subsystem test does not satisfy a vehicle-level requirement unless its final hardware, configuration, and interfaces are represented.

10.2 Configuration Control and Pre-Test Record

Every trial should be associated with a configuration record containing the repository commit, firmware build flags, parameter-file checksum, controller board revision, PCB revision, actuator firmware and operating mode, wheel order and handedness, battery identification and state of charge, payload, tire condition, and raw-data filename. Configuration control is necessary because a sign change, node reassignment, filter coefficient, or current limit can materially change the outcome.

<VERIFY FROM HARDWARE: Confirm the installed motor model and quantity; firmware version; actuator command mode; four CAN node IDs; CAN bitrate; current, voltage, speed, temperature, and ERPM limits; encoder zero convention; wheel-direction signs; radio failsafe values; controller build identifier; and final PCB revision before testing.>

Table 10.2. Test instrumentation and measurement channels

Quantity	Instrument/channel	Range/resolution	Rate	Calibration/traceability
Pitch and pitch rate	Onboard IMU plus independent reference	<INSERT>	<INSERT Hz>	Six-position accelerometer and stationary gyro tests
Wheel position/speed	Four actuator feedback channels	<INSERT>	<INSERT Hz>	Direction and counts/revolution verification
Motor current	Controller feedback plus external sensor	<INSERT A>	<INSERT Hz>	Compare at two or more known loads
Battery voltage	Onboard ADC/BMS plus DMM	<INSERT V>	<INSERT Hz>	Two-point comparison
Body pose	Motion capture, camera, or surveyed floor markers	<INSERT>	<INSERT Hz>	Scale and coordinate-frame calibration
Temperatures	Actuator/controller/battery sensors	<INSERT °C>	<INSERT Hz>	Ambient and reference comparison
Loop timing/CAN	Firmware time stamps and bus analyzer	<INSERT μ s>	Per event	Common time base or measured offset

<INSERT FIGURE 10.2: Show a bench power supply or protected battery connection, current limit, fuses, emergency disconnect, controller, CAN analyzer/oscilloscope, four actuators, termination points, and measurement references. Label expected voltage domains and the safe operator position>
 What the figure should show: Show a bench power supply or protected battery connection, current limit, fuses, emergency disconnect, controller, CAN analyzer/oscilloscope, four actuators, termination points, and measurement references. Label expected voltage domains and the safe operator position
 Likely source: laboratory photograph and annotated electrical diagram

Figure 10.2. Current-limited electrical bring-up and CAN validation setup.

10.3 Power-On and Communication Validation

With the wheels unloaded and propulsion disabled, the proposed procedure begins with resistance and polarity checks. Logic rails are energized from a current-limited source before actuator power is connected. Rail voltage, ripple, regulator temperature, and idle current are recorded. CANH and CANL resistance is measured with power removed; approximately 60 Ω is expected only when two 120 Ω end terminations are present in parallel [25]. With power applied, recessive and dominant levels, edge integrity, error counters, and traffic load are observed. Each motor is then enabled individually at a low command while its physical direction and feedback sign are recorded.

Abort criteria include reversed polarity, unexpected current, regulator dropout, repeated CAN error-passive or bus-off transitions, actuator faults, uncontrolled motion, excessive temperature, stale feedback, or disagreement between commanded and observed direction. The emergency disconnect must be reachable without crossing the robot's fall envelope.

10.4 Sensor Calibration and Coordinate-Frame Verification

The final IMU installation will be calibrated after the mounting hardware is fixed. Accelerometer offset and scale are estimated using six static orientations; gyroscope bias and noise are estimated from a stationary record after thermal stabilization. The body-to-sensor rotation is then verified by applying positive roll, pitch, and yaw rotations one at a time. A forward chassis tilt must produce the pitch sign assumed by the controller, and a positive wheel command must produce the modeled body-force sign. Independent observation is required because an internally consistent but inverted sensor-and-actuator convention can pass software unit tests while destabilizing the physical robot.

Table 10.3. Calibration record

Item	Method	Raw record	Result	Date/configuration
Accelerometer bias/scale	Six static orientations	<INSERT FILE>	<INSERT vector/matrix>	<INSERT>
Gyro bias/noise	Stationary thermal record	<INSERT FILE>	<INSERT rad/s and noise>	<INSERT>
IMU mounting rotation	Single-axis sign tests	<INSERT FILE>	<INSERT transform>	<INSERT>
Wheel direction and feedback	Lifted low-speed command	<INSERT FILE>	<INSERT four signs>	<INSERT>
Effective radius	Loaded rollout	<INSERT FILE>	<INSERT m $\pm$ uncertainty>	<INSERT>
Current and voltage channels	Reference instrument comparison	<INSERT FILE>	<INSERT calibration>	<INSERT>

10.5 Restrained Controller Commissioning

Initial balance trials will use a low-friction restraint or pivot that prevents a fall from damaging equipment without carrying normal balance torque. The system will start in DISARMED, verify fresh IMU and actuator feedback, confirm a valid receiver state, and require an explicit neutral arming command. The first closed-loop test will use conservative torque and slew limits with integral action disabled. Feedback polarity will be checked using a small manual displacement while the wheels are clear of the ground, followed by low-authority ground contact. Controller bandwidth will be increased only after delay, noise, saturation, and structural compliance have been measured.

<INSERT FIGURE 10.3: Depict the restraint geometry, allowed pitch motion, load-free directions, emergency stop, operator position, protected fall envelope, and any tether forces. The restraint must not create an unmodeled restoring moment over the tuning range>

What the figure should show: Depict the restraint geometry, allowed pitch motion, load-free directions, emergency stop, operator position, protected fall envelope, and any tether forces. The restraint must not create an unmodeled restoring moment over the tuning range

Likely source: CAD drawing and laboratory photograph of the commissioning fixture

Figure 10.3. Mechanical restraint used for initial balance-controller tuning.

10.6 Free-Balance and Disturbance-Rejection Tests

After restrained stability is demonstrated, free-balance trials will be performed within a clear padded area. A test begins from a documented pitch angle and neutral planar command. At least five successful repeats and all unsuccessful attempts will be retained. Pitch settling time, percent overshoot where meaningful, steady-state bias, RMS error over a stated interval, peak command, saturation duration, and feedback age are computed using fixed definitions. Disturbance rejection will use a repeatable impulse mechanism or a controlled initial displacement; an undocumented hand push is acceptable only as a qualitative demonstration and not as the principal quantitative result.

<INSERT DATA: Measured pitch-angle response during recovery from an approximately [ANGLE]-degree initial displacement. Required signals: pitch angle (deg), pitch rate (deg/s), four motor commands and feedback values, saturation flags, operating state, IMU age, CAN-feedback age, battery voltage (V), and time (s). Provide at least five trials and raw machine-readable files.>

10.7 Longitudinal, Lateral, and Yaw Tests

Planar tests will be introduced sequentially while the balance loop remains active. Step and trapezoidal references will be used for longitudinal velocity, lateral velocity, and yaw rate, followed by combined commands. Reference magnitudes will be bounded by measured torque,

speed, traction, and balance reserves. External pose measurement is preferred because wheel integration cannot distinguish true lateral motion from roller slip. Each trial will report reference and measured motion, pitch response, all wheel commands, current, saturation, and the allocation residual.

The collinear geometry warrants an additional identifiability check: pure lateral and pure yaw commands must generate measurably distinct wheel-speed patterns and body responses. If H is poorly conditioned or the assembled roller pattern is rank deficient, the affected degree of freedom will be reported as constrained rather than hidden through controller tuning.

10.8 Power, Thermal, Endurance, and Fault Tests

A staged endurance test will log battery voltage, branch current, regulator voltage and temperature, actuator temperatures, CAN errors, loop overruns, and fault state while the robot executes a repeatable bounded maneuver. Test duration and ambient conditions must be stated. Separate injected-fault tests will remove receiver updates, delay motor feedback, disconnect one actuator while safe, exceed the permitted tilt, and approach undervoltage thresholds. The expected response is a deterministic transition to a documented safe state, not merely cessation of logging. Battery testing will follow the battery manufacturer's handling and discharge limits.

<INSERT FIGURE 10.4: Show the shared time base among firmware logs, CAN analyzer, external pose or video, current/voltage instrumentation, and thermal sensors. Identify trigger events, measured synchronization error, sample rates, and file naming>

What the figure should show: Show the shared time base among firmware logs, CAN analyzer, external pose or video, current/voltage instrumentation, and thermal sensors. Identify trigger events, measured synchronization error, sample rates, and file naming

Likely source: data-acquisition schematic and photograph

Figure 10.4. Synchronized data-acquisition and reference-measurement arrangement.

10.9 Repeatability, Uncertainty, and Data Reduction

Raw data will remain immutable and will be processed by version-controlled scripts. Time stamps will be converted to seconds on a common origin, duplicate or missing samples will be reported, and filtering will be applied only once with documented order, cutoff, and phase characteristics. Uncertainty will combine sensor calibration, time alignment, repeatability, and reference-system resolution where applicable. Means, standard deviations, confidence intervals, and trial counts will accompany aggregate metrics. Failed trials are part of the evidence and will not be discarded without a stated exclusion rule.

<INSERT PROCEDURE: Provide the final laboratory risk assessment, approved battery-charging and storage procedure, emergency-stop design, test-area layout, required personal protective equipment, maximum authorized command/current/tilt limits, abort criteria, and names of trained observers.>

Chapter 11

RESULTS AND DISCUSSION

At the date of this first draft, the repository substantiates subsystem development but does not contain a complete, traceable data package demonstrating upright balance, disturbance rejection, or three-degree-of-freedom planar motion. The chapter therefore reports the defensible qualitative results and provides a fixed structure for quantitative results. Bracketed result fields must be populated from raw data before submission. Statements that depend on those fields are explicitly conditional.

11.1 Confirmed Development Results

The project has established a nonconventional mechanical concept: four Mecanum wheels share a common transverse line rather than occupying the corners of a rectangle. Four integrated CubeMars actuators and 104 mm nominal Mecanum wheels were selected. An ESP32 firmware implementation partitions portable interfaces and device drivers from board-specific tasks, acquires the ISM330DHCX, performs mounting and gyro calibration steps, produces a complementary pitch estimate, receives CRSF commands, updates a local display, communicates with actuators through TWAI/CAN, monitors feedback and faults, and gates commands through supervisor logic [9]. These are integration milestones; none independently proves closed-loop vehicle performance.

The electrical repository includes a KiCad project and a developing power/control design. Component selection and schematic work are more mature than physical PCB validation. The present evidence does not establish a fabricated and tested custom board, final protection design, conducted or radiated compatibility, or continuous-current capability. Similarly, the mathematical development establishes a symbolic framework and a rank test, but numerical predictions remain conditional on measured geometry and mass properties.

11.2 Balancing Performance

The primary balance result will be evaluated over a predefined interval beginning after arming or disturbance release. Let $e_{\theta}[k] = \theta_{\text{ref}}[k] - \theta[k]$. The RMS error is the square root of the mean of e_{θ}^2 over N valid samples. Settling time is the earliest time after which the absolute pitch error remains within a stated band for the remainder of the evaluation interval. Overshoot is reported only when the response and reference make that quantity well defined. Saturation time, fall-state transitions, and excluded samples are reported alongside error metrics.

$$\theta_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{k=1}^N (\theta_k - \theta_{ref,k})^2} \quad (11.1)$$

Table 11.1. Balancing and motion-performance results

Metric	Simulation	Hardware mean $\pm$ SD	Trials	Acceptance criterion	Status
RMS pitch error (deg)	<INSERT>	<INSERT>	<INSERT N>	<AUTHOR CONFIRM>	Pending
Peak pitch excursion (deg)	<INSERT>	<INSERT>	<INSERT N>	<AUTHOR CONFIRM>	Pending
Settling time (s)	<INSERT>	<INSERT>	<INSERT N>	<AUTHOR CONFIRM>	Pending
Steady-state pitch bias (deg)	<INSERT>	<INSERT>	<INSERT N>	<AUTHOR CONFIRM>	Pending
Maximum recoverable initial angle (deg)	<INSERT>	<INSERT>	<INSERT N>	<AUTHOR CONFIRM>	Pending
Longitudinal velocity RMSE (m/s)	<INSERT>	<INSERT>	<INSERT N>	<AUTHOR CONFIRM>	Pending
Lateral velocity RMSE (m/s)	<INSERT>	<INSERT>	<INSERT N>	<AUTHOR CONFIRM>	Pending
Yaw-rate RMSE (rad/s)	<INSERT>	<INSERT>	<INSERT N>	<AUTHOR CONFIRM>	Pending
Peak current per branch (A)	<INSERT>	<INSERT>	<INSERT N>	Hardware limit	Pending
Worst control-cycle execution (μ s)	<INSERT>	<INSERT>	<INSERT N>	Below period	Pending

<INSERT FIGURE 11.1: Plot time (s) horizontally. Use aligned panels for pitch reference and measured pitch (deg), pitch rate (deg/s), collective balance command and individual actuator commands (documented units), saturation/fault flags, and operating state. Show all trials faintly and the mean with uncertainty; identify the metric interval and controller configuration>

What the figure should show: Plot time (s) horizontally. Use aligned panels for pitch reference and measured pitch (deg), pitch rate (deg/s), collective balance command and individual actuator commands (documented units), saturation/fault flags, and operating state. Show all trials faintly and the mean with uncertainty; identify the metric interval and controller configuration

Likely source: synchronized raw balance-test logs processed by a version-controlled script

Figure 11.1. Measured pitch response during upright balancing.

<INSERT RESULT: RMS pitch error, maximum pitch excursion, settling time, overshoot, steady-state error, saturation duration, and number of successful/attempted trials from the completed balancing experiment. State the calculation interval and uncertainty.>

11.3 Disturbance Rejection

Disturbance trials will be grouped by applied impulse or initial angle rather than pooled indiscriminately. The discussion will distinguish recovery without saturation, recovery with saturation, transition to FALLEN, and mechanical restraint contact. A stronger conclusion—that the robot rejects disturbances up to a specified magnitude—will be made only after the disturbance input is measured or repeatably defined.

<INSERT FIGURE 11.2: Plot time from disturbance (s) versus pitch angle (deg), pitch rate (deg/s), applied or proxy disturbance, four actuator commands, battery voltage (V), and operating state. Compare at least five repetitions at each disturbance level and mark recovery/fall classification>

What the figure should show: Plot time from disturbance (s) versus pitch angle (deg), pitch rate (deg/s), applied or proxy disturbance, four actuator commands, battery voltage (V), and operating state. Compare at least five repetitions at each disturbance level and mark recovery/fall classification

Likely source: instrumented pendulum/impulse test or controlled initial-displacement logs

Figure 11.2. Measured recovery from repeatable pitch disturbances.

11.4 Planar Motion and Mobility

Longitudinal, lateral, and yaw tests will first be interpreted independently. Reference-to-measurement delay, steady-state gain, RMS error, cross-axis response, pitch deviation, and actuator effort will be reported. Pure lateral response is especially important: wheel rotation alone is insufficient evidence because roller deformation and slip can generate large odometric motion without equivalent chassis translation. External pose data are therefore required for a quantitative lateral-mobility claim.

<INSERT FIGURE 11.3: Create panels for longitudinal velocity v_x (m/s), lateral velocity v_y (m/s), yaw rate (rad/s), x-y trajectory (m), pitch (deg), four wheel speeds (rad/s), and allocation residual. Compare reference, external measurement, wheel-based estimate, and simulation at a common sample interval>
 What the figure should show: Create panels for longitudinal velocity v_x (m/s), lateral velocity v_y (m/s), yaw rate (rad/s), x-y trajectory (m), pitch (deg), four wheel speeds (rad/s), and allocation residual. Compare reference, external measurement, wheel-based estimate, and simulation at a common sample interval
 Likely source: motion-capture/camera data synchronized with firmware and simulation logs

Figure 11.3. Measured longitudinal, lateral, and yaw responses.

<INSERT RESULT: For separate and combined commands, report longitudinal-velocity RMSE, lateral-velocity RMSE, yaw-rate RMSE, cross-axis coupling, trajectory endpoint error, maximum pitch deviation, allocation residual, and saturation fraction.>

11.5 Electrical, Thermal, Communication, and Timing Results

Vehicle performance depends on electrical and computational margins. Voltage sag can reduce actuator authority, temperature can trigger derating, stale CAN feedback can invalidate the state used by the controller, and scheduling jitter can reduce phase margin. The final analysis will place these signals on the same time axis as pitch and commands. Datasheet limits will be identified separately from project-configured limits and measured values.

<INSERT FIGURE 11.4: Plot time (s) versus battery and regulated-rail voltage (V), total and per-branch current (A), cumulative energy (Wh), actuator/controller/battery temperature ($^{\circ}\text{C}$), and commanded operating state. Include ambient temperature and limit lines from the verified configuration>
 What the figure should show: Plot time (s) versus battery and regulated-rail voltage (V), total and per-branch current (A), cumulative energy (Wh), actuator/controller/battery temperature ($^{\circ}\text{C}$), and commanded operating state. Include ambient temperature and limit lines from the verified configuration
 Likely source: power analyzer, firmware telemetry, and calibrated temperature logs

Figure 11.4. Motor current, battery voltage, and actuator temperature during testing.

<INSERT FIGURE 11.5: Plot histograms and time histories of control-cycle period and execution time (μs), IMU sample age (μs), motor-feedback age (μs), CAN utilization (%), error frames, retries/timeouts, and task overruns. Report median, 95th, 99th, and worst-case values>
 What the figure should show: Plot histograms and time histories of control-cycle period and execution time (μs), IMU sample age (μs), motor-feedback age (μs), CAN utilization (%), error frames, retries/timeouts, and task overruns. Report median, 95th, 99th, and worst-case values
 Likely source: instrumented firmware time stamps and CAN analyzer capture

Figure 11.5. Measured control-loop timing and CAN communication performance.

11.6 Requirement Verification

Table 11.2. Requirement-verification results

Requirement	Evidence required	Observed result	Pass criterion	Verdict
R-F01 controlled upright balance	Free-balance logs	<INSERT>	Approved pitch/settling limits	Pending
R-F02 commanded planar motion	External pose and commands	<INSERT>	Approved tracking limits	Pending
R-C01 rank-adequate allocation	Measured H and singular values	<INSERT>	rank(H)=3 with acceptable conditioning	Pending
R-E01 protected power distribution	Schematic/layout/bench test	<INSERT>	No limit violation or fault	Pending
R-S01 safe loss-of-link response	Injected timeout test	<INSERT>	Deterministic safe transition	Pending
R-S02 stale-feedback response	Injected delay/dropout	<INSERT>	Commands inhibited within limit	Pending
R-P01 timing	Timestamp distribution	<INSERT>	Worst case below approved deadline	Pending
R-M01 serviceability	Inspection/replacement demonstration	<INSERT>	Approved procedure/time	Pending

11.7 Failure Cases and Limitations

The expected failure modes include incorrect IMU or wheel signs, estimator drift or vibration sensitivity, actuator saturation, feedback latency, CAN node misconfiguration, voltage sag, tire/roller slip, compliance, insufficient yaw leverage, a rank-deficient roller arrangement, and a center of mass outside the modeled range. A result is not removed merely because it is unfavorable; the associated configuration and evidence are retained so the limitation can guide redesign.

Table 11.3. Failure and anomaly log

Event/time	Configuration	Observed symptom	Evidence	Root cause	Corrective action/retest
<INSERT>	<COMMIT/BUILD>	<INSERT>	<LOG/PHOTO>	<CONFIRMED OR HYPOTHESIS>	<INSERT>
<INSERT>	<COMMIT/BUILD>	<INSERT>	<LOG/PHOTO>	<CONFIRMED OR HYPOTHESIS>	<INSERT>
<INSERT>	<COMMIT/BUILD>	<INSERT>	<LOG/PHOTO>	<CONFIRMED OR HYPOTHESIS>	<INSERT>

The present evidence supports the conclusion that Zephyr is a substantially developed mechatronic prototype with operational subsystems and a coherent path to validation. It does not yet support the stronger conclusions that the robot balances autonomously, achieves holonomic motion, meets quantitative requirements, or is electrically and thermally qualified. Those conclusions remain pending the tests and result fields in this chapter.

Chapter 12

CONCLUSIONS AND FUTURE WORK

12.1 Conclusions

This thesis has documented the design and development state of Zephyr, a four-actuator mobile robot whose Mecanum wheels are placed along a common axis and whose narrow support geometry requires active pitch stabilization. The work unifies mechanical configuration, electrical architecture, embedded firmware, geometry-parameterized kinematics, inverted-pendulum dynamics, control planning, and a traceable validation program. The formulation intentionally avoids applying the familiar rectangular four-corner Mecanum matrix to a geometry for which its lever arms and roller ordering have not been verified.

Repository evidence confirms meaningful subsystem progress. Four AK40-10 actuators and 104 mm nominal Mecanum wheels have been selected; the mechanical prototype is substantially assembled; IMU, receiver, display, and CAN actuator interfaces operate in the ESP32 development firmware; estimation and supervisor infrastructure exist; and a custom electrical design is in development. A symbolic Jacobian, mobility-rank test, nonlinear balance model, upright linearization, and constrained allocation formulation provide the analytical basis for controller design.

The central research question—whether the assembled collinear mechanism can maintain balance while producing independently controlled longitudinal, lateral, and yaw motion—cannot yet be answered quantitatively. The exact wheel sequence, roller signs, wheel coordinates, mass properties, actuator configuration, final control gains, timing distribution, and synchronized test data remain required. No claim of stable balance, holonomic mobility, disturbance tolerance, tracking accuracy, power margin, or thermal margin is made in their absence.

Table 12.1. Objective completion at the time of this draft

Objective	Evidence completed	Status	Evidence still required
Document actual architecture	Repository-based subsystem and interface record	Partially achieved	Final as-built survey and branch-complete archive
Establish collinear kinematics	Symbolic geometry/sign model and rank test	Analytically achieved	Measured y_i , s_i , radius, and experimental validation
Model balance dynamics	Nonlinear and linear symbolic models	Analytically achieved	Measured mass properties and identified losses
Implement embedded infrastructure	IMU, CRSF, display, CAN, feedback, supervisor paths	Substantially achieved	Release build, timing, four-node configuration, logging
Implement stabilizing controller	Architecture and safety constraints specified	Not yet demonstrated	Code, gains, constrained test logs
Demonstrate omnidirectional motion while balancing	Test method specified	Not yet demonstrated	External pose and synchronized vehicle data

12.2 Mechanical Future Work

The highest-priority mechanical task is an as-built survey. It should record wheel-center coordinates, wheel order, roller handedness, effective loaded radius, frame dimensions, fasteners, shafts, bearings, couplings, actuator mount stiffness, component masses, complete vehicle mass, center of mass, and principal inertias. CAD should then be reconciled with the assembled prototype. The design should provide repeatable wheel alignment, protected cable routing, accessible fasteners, a battery restraint, lifting points, and a fall structure that protects hands, electronics, and rollers without interfering with normal motion. Structural calculations and safety factors should be added only after materials, loads, and interfaces are confirmed.

12.3 Electrical Future Work

The schematic must be frozen only after the installed battery, protection approach, connectors, fuse ratings, regulator range, CAN transceiver, termination, and actuator peak demands are verified. The PCB should be completed with a documented copper weight, stack-up, branch-current assumptions, thermal paths, return-current control, decoupling, transient protection, and design-rule review. Low-voltage bring-up should precede battery testing, and the fabricated assembly should undergo short-circuit inspection, rail-load testing, thermal testing, CAN signal-integrity inspection, undervoltage response, and fault-injection tests.

12.4 Firmware and Control Future Work

Firmware priorities are to freeze a reproducible ESP32 baseline, verify all four CAN node IDs and directions, measure task timing and feedback age, implement structured logging, and complete the command-to-actuator allocation. The controller should first be validated in restrained tests with explicit saturation, rate limits, anti-windup, stale-data rejection, tilt limits, and receiver failsafe. LQR or another state-feedback law may be used for the inner balance loop if the measured model is controllable and the estimator supplies reliable states [13]; outer planar loops should be added only after balance torque reserve is quantified. The STM32H723 migration should remain a separate milestone until behavior is equivalent on the present platform.

12.5 Experimental Future Work

The proposed experiment sequence is: as-built metrology; power and CAN validation; IMU and actuator calibration; restrained balance; free balance; repeatable disturbance recovery;

independent longitudinal, lateral, and yaw commands; combined trajectories; and endurance and fault injection. Each stage should have approved quantitative requirements and stop criteria before execution. External pose measurement, independent electrical instrumentation, synchronized clocks, repeated trials, uncertainty reporting, and retention of failures are needed for a defensible model-to-hardware comparison.

When these steps are complete, the final thesis should replace the provisional abstract, pending result language, and ready-to-fill tables with measured values. The strongest potential contribution is not simply a four-wheel robot, but a validated account of the mobility and control consequences of placing four Mecanum wheels on one line while simultaneously stabilizing an inverted body. Whether that potential is realized remains an empirical question to be answered by the documented test program.

REFERENCES

- [1] F. Grasser, A. D'Arrigo, S. Colombi, and A. C. Rufer, "JOE: A mobile, inverted pendulum," *IEEE Transactions on Industrial Electronics*, vol. 49, pp. 107–114, Feb. 2002.
- [2] K. Pathak, J. Franch, and S. K. Agrawal, "Velocity and position control of a wheeled inverted pendulum by partial feedback linearization," *IEEE Transactions on Robotics*, vol. 21, pp. 505–513, June 2005.
- [3] F. G. Pin and S. M. Killough, "A new family of omnidirectional and holonomic wheeled platforms for mobile robots," *IEEE Transactions on Robotics and Automation*, vol. 10, pp. 480–489, Aug. 1994.
- [4] B. E. Ilon, "Wheels for a course stable selfpropelling vehicle movable in any desired direction on the ground or some other base." U.S. Patent 3,876,255, Apr. 1975.
- [5] A. Gferrer, "Geometry and kinematics of the mecanum wheel," *Computer Aided Geometric Design*, vol. 25, pp. 784–791, Dec. 2008.
- [6] G. Indiveri, "Swedish wheeled omnidirectional mobile robots: Kinematics analysis and control," *IEEE Transactions on Robotics*, vol. 25, pp. 164–171, Feb. 2009.
- [7] L. Ferrière, B. Raucent, and G. Campion, "Design of omnimobile robot wheels," in *Proceedings of the IEEE International Conference on Robotics and Automation*, (Minneapolis, MN, USA), pp. 3664–3670, 1996.
- [8] P. F. Muir and C. P. Neuman, "Kinematic modeling of wheeled mobile robots," *Journal of Robotic Systems*, vol. 4, no. 2, pp. 281–340, 1987.
- [9] A. Kyaw, "e_me599_Zephyr: Collinear mecanum-drive self-balancing mobile robot." GitHub repository. Accessed: Sep. 18, 2026.
- [10] G. Bayar and S. Ozturk, "Investigation of the effects of contact forces acting on rollers of a mecanum wheeled robot," *Mechatronics*, vol. 72, p. 102467, Dec. 2020.
- [11] A. Bárcenas-Presteguí, R. A. García-García, A. G. Loukianov, and E. N. Sanchez, "Robust control of an omnidirectional self-balancing robot," *Control Theory and Technology*, vol. 23, pp. 410–422, 2025.
- [12] CubeMars, "AK40-10 robotic actuator." Manufacturer product specifications; accessed Sep. 18, 2026.
- [13] B. D. O. Anderson and J. B. Moore, *Optimal Control: Linear Quadratic Methods*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1990.

- [14] H. K. Khalil, *Nonlinear Systems*. Upper Saddle River, NJ, USA: Prentice Hall, 3 ed., 2002.
- [15] S. Chacko and S. Abraham, “Control strategies for trajectory tracking of a two-wheeled self-balancing robot,” *International Journal of Dynamics and Control*, vol. 11, pp. 1584–1592, 2023.
- [16] A. Bhargava, M. Suhaib, and A. K. Singholi, “Modeling and control of four mecanum wheeled mobile robot: A systematic review,” *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, vol. 48, p. 570, 2026.
- [17] G. Campion, G. Bastin, and B. D’Andréa-Novel, “Structural properties and classification of kinematic and dynamic models of wheeled mobile robots,” *IEEE Transactions on Robotics and Automation*, vol. 12, pp. 47–62, Feb. 1996.
- [18] C. L. Liu and J. W. Layland, “Scheduling algorithms for multiprogramming in a hard-real-time environment,” *Journal of the ACM*, vol. 20, pp. 46–61, Jan. 1973.
- [19] K. J. Åström and R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton, NJ, USA: Princeton University Press, 2008.
- [20] R. Mahony, T. Hamel, and J.-M. Pflimlin, “Nonlinear complementary filters on the special orthogonal group,” *IEEE Transactions on Automatic Control*, vol. 53, pp. 1203–1218, June 2008.
- [21] Espressif Systems, *ESP32 Series Datasheet*, version 5.2 ed. Accessed: Sep. 18, 2026.
- [22] Espressif Systems, *Two-Wire Automotive Interface (TWAI), ESP-IDF Programming Guide*. Accessed: Sep. 18, 2026.
- [23] NXP Semiconductors, *TJA1051 High-Speed CAN Transceiver*. Accessed: Sep. 18, 2026.
- [24] Adafruit Industries, *Adafruit CAN Pal*. Accessed: Sep. 18, 2026.
- [25] Texas Instruments, *Automotive Daisy-Chain CAN Bus Transceiver Reference Design*. Accessed: Sep. 18, 2026.
- [26] FreeRTOS, *Task Scheduling*. FreeRTOS kernel documentation; accessed Sep. 18, 2026.
- [27] STMicroelectronics, *STM32H723xE/G 32-bit Arm Cortex-M7 MCU*. Accessed: Sep. 18, 2026.
- [28] goBILDA, “Gripforce mecanum wheel set (ø104 mm, 40a durometer rollers).” Manufacturer product specifications; accessed Sep. 18, 2026.
- [29] Texas Instruments, *BQ76942 3-Series to 10-Series High-Accuracy Battery Monitor and Protector*. Accessed: Sep. 18, 2026.

- [30] STMicroelectronics, *ISM330DHCX: iNEMO Inertial Module*. Accessed: Sep. 18, 2026.
- [31] RadioMaster, “User manuals.” Accessed: Sep. 18, 2026.
- [32] Waveshare, “2inch LCD module.” Manufacturer wiki; accessed Sep. 18, 2026.
- [33] Analog Devices, *LT8609S: 42 V, 2 A/3 A Peak Synchronous Step-Down Regulator*. Accessed: Sep. 18, 2026.
- [34] IPC, “Standard for determining current-carrying capacity in printed board design.” IPC-2152, Aug. 2009.
- [35] ExpressLRS Project, “CRSF protocol definitions.” Source repository; accessed Sep. 18, 2026.

Chapter A

SYSTEM PARAMETERS

Table A.1 is the single source of truth for numerical parameters used in the final model, controller, simulation, and results. Each value should carry a method, uncertainty, configuration, and source. A datasheet value must not silently replace an as-installed or experimentally identified value.

Table A.1. Master system-parameter record

Category	Parameter/symbol	Value	SI unit	Method/source	Uncertainty/status
Geometry	Effective wheel radius, r	<INSERT>	m	Loaded rollout; compare CAD	<INSERT>
Geometry	Wheel coordinates, (x_i, y_i)	<INSERT 4 ROWS>	m	As-built survey	Required
Geometry	Roller signs, s_i	<INSERT 4 SIGNS>	—	Visual and motion sign test	Required
Geometry	Roller angle	<VERIFY>	rad	Wheel drawing/measurement	Required
Mass	Complete robot mass	<INSERT>	kg	Calibrated scale	Required
Mass	Body mass above axis, m_b	<INSERT>	kg	Component/CAD breakdown	Required
Mass	COM vector from balance axis	<INSERT>	m	CAD plus physical check	Required
Inertia	Pitch/yaw inertias	<INSERT>	kg·m ²	CAD or system identification	Required
Actuator	Model and quantity	AK40-10; four selected	—	Repository/product record [9], [12]	Verify installed labels
Actuator	Gear ratio	10:1 nominal	—	Manufacturer [12]	Verify revision
Actuator	Rated/peak output torque	1.3 / 4.1 nominal	N·m	Manufacturer [12]	Rating; not project limit
Actuator	Rated/no-load output speed	370 / 435 nominal	rpm	Manufacturer [12]	Rating; verify voltage
Actuator	Command mode and scaling	<INSERT>	mode-specific	Installed firmware/config	Required
Actuator	Four node IDs and directions	Motor 1 observed as 69; others <INSERT>	—	Firmware/hardware survey	Incomplete
Electrical	Battery chemistry/cell count	LiPo, 6S reported	—	Label and configuration	Verify pack
Electrical	Battery nominal/max voltage	<INSERT VERIFIED>	V	Exact pack datasheet/label	Required
Electrical	Capacity/discharge rating	<INSERT>	A·h / C	Exact pack label	Required
Sensor	IMU	ISM330DHCX	—	Driver/configuration [9], [30]	Confirmed in code; verify board
Sensor	Accel/gyro ranges	± 4 g / ± 500 deg/s observed	m/s ² / rad/s	Firmware plus datasheet	Verify final build
Sensor	Output data rate	208 observed	Hz	Firmware plus datasheet	Verify final build
Comms	CAN bitrate	1,000,000 observed	bit/s	ESP32 configuration	Verify all nodes
Control	Estimator/controller period	<INSERT MEASURED>	s	Firmware timestamps	Required
Control	Feedback age limit	<INSERT>	s	Safety configuration	Required
Control	Gain matrices/limits	<INSERT>	various	Released parameter file	Not finalized
Losses	b_x , b_{θ} /traction terms	<INSERT>	SI	System identification	Required for prediction

Chapter B

PROJECT EVIDENCE AND COMPLETENESS REVIEW

The inventory below is the evidence boundary used for this first draft. ‘Confirmed’ means directly supported by accessible repository content or an official project record; it does not imply system-level validation. Confidence reflects the quality and specificity of the evidence, not confidence that a subsystem meets an unstated performance target.

Table B.1. Project evidence and completeness inventory

Subsystem	Apparent status	Supporting repository paths/evidence	Missing information	Confidence
Concept/architecture	Implemented in prototype; integration continuing	README.md; 04_firmware; project records	Final as-built block diagram and configuration	High
Mechanical frame	Substantially constructed; exact geometry unavailable	README imagery/project records; no verified CAD path in accessible root	CAD, dimensions, materials, joints, masses	Medium
Four-wheel collinear layout	Project-defining configuration	README.md description and project title	Wheel order, y_i, roller signs, photographs	High concept / low geometry
Wheels	Selected; installation to verify	Project record; manufacturer data [28]	Installed handedness, effective radius, wear	Medium-high
Actuators	Four selected; motion achieved during bring-up	04_firmware configuration; manufacturer [12]	Four IDs, modes, directions, limits, firmware	Medium-high
Battery/power	6S LiPo concept; board design developing	03_electrical/Zephyr.kicad_sch; project records	Exact pack, fusing, BMS/protection, test results	Medium
Custom PCB	Schematic/layout work in progress; fabrication unconfirmed	03_electrical/Zephyr.kicad_pro, .kicad_sch, .kicad_pcb	Final review, routing, fabrication, bring-up	High status
Controller	ESP32 active; STM32H723 target/scaffold	04_firmware/platforms/esp32; platforms/stm32; archive	Exact board revision and migration decision	High
IMU/estimation	Driver, transform, calibration, complementary estimator implemented	common/include/drivers/drv_ism330dhcx.h; common/src/drivers/drv_ism330dhcx.c; control types	Final mounting matrix, calibration logs, estimator validation	High code / medium hardware
CAN interface	ESP32 TWAI communication and feedback/fault monitoring reported	platforms/esp32; config/config_actuator.h; Adafruit CAN Pal [24]	Four-node capture, termination, harness map, error statistics	Medium-high
Receiver/display	CRSF and ST7789 operation reported	platforms/esp32 configuration/modules	Final pin/harness verification and failsafe test	Medium-high
Safety/supervisor	Infrastructure present; vehicle-level validation absent	04_firmware common/platform supervisor paths	State-transition tests, E-stop, hard power isolation	Medium
Balance controller	Architecture/types present; closed-loop implementation not confirmed	common/include/control/ctrl_balance_types.h	Control law, gains, discrete timing, hardware evidence	High status
Wheel allocation	Derived in thesis; operational code not confirmed	No verified current allocation module	Measured H, constrained allocator, unit tests	High status
Simulation	05_simulation now visible at root; contents not traversable; earlier MATLAB artifact not accepted as repository result	05_simulation; prior project artifact record	Complete scripts, parameters, raw outputs, README	Low-current
Experimental validation	No traceable integrated data package located	No verified results/log directory	Raw logs, procedures, configurations, plots, trial counts	High status
Version history	Main root reports 54 commits; full history/branches inaccessible in this session	GitHub repository landing page	git bundle or ZIP plus git log --all	Medium

Table B.2. Proposed thesis titles

Designation	Title	Rationale
Proposed	Design and Development of a Self-Balancing Mobile Robot with a Collinear Four-Wheel Mecanum Drive	Accurately names design/development, balance function, and nonstandard collinear drive.
Alternative 1	Design, Modeling, and Embedded Control of Zephyr: A Collinear Four-Wheel Mecanum Balancing Robot	Emphasizes modeling and embedded implementation.
Alternative 2	Mechatronic Development of a Dynamically Balanced Omnidirectional Robot with Four Collinear Mecanum Wheels	Highlights system integration and intended mobility without claiming validated holonomy.
Alternative 3	Architecture and Validation Framework for a Collinear Mecanum-Wheel Inverted-Pendulum Robot	Best if final experimental results remain limited; emphasizes defensible completed contribution.

Chapter C

PIN, WIRING, AND CONNECTOR TABLES

The pin maps below reproduce the active ESP32 prototype evidence available during drafting. They are not a wiring authorization. Crossed transmit/receive naming, board silkscreen, logic voltage, connector orientation, and ground reference must be verified at the installed hardware before energization.

Table C.1. ESP32 prototype pin and interface record

Interface	Signal	ESP32 GPIO	Voltage domain	Direction at MCU	Verification
ISM330DHCX SPI	SCLK	18	3.3 V	Out	Observed configuration; verify
ISM330DHCX SPI	MISO/SDO	19	3.3 V	In	Observed configuration; verify
ISM330DHCX SPI	MOSI/SDI	23	3.3 V	Out	Observed configuration; verify
ISM330DHCX SPI	CS	27	3.3 V	Out	Observed configuration; verify
ISM330DHCX	INT1 / INT2	34 / 35	3.3 V	In	Observed configuration; verify use
ST7789 display	DIN / CLK	13 / 14	3.3 V	Out	Observed configuration; verify
ST7789 display	CS / DC / RST	25 / 21 / 22	3.3 V	Out	Observed configuration; verify
CAN Pal	TX / RX	26 / 32	3.3 V	Out / In	Observed configuration; verify
RP2 receiver	MCU RX / MCU TX	16 / 17	UART logic; receiver power 5 V	In / Out	Verify crossed harness and logic levels

Table C.2. Power and CAN connector record

Connector/net	Pins/conductors	Source	Destination	Wire/rating	Protection/notes
Battery input A	<INSERT>	6S pack/branch	Motors <INSERT>	14 AWG recommended; verify	Independent fuse; polarity keying
Battery input B	<INSERT>	6S pack/branch	Motors <INSERT>	14 AWG recommended; verify	Independent fuse; polarity keying
Logic regulator input	<INSERT>	Protected battery bus	LT8609S stage	<INSERT>	Reverse/transient/overcurrent provisions
J9–J12 CAN	Pin 1 CANH; pin 2 CANL observed	Linear CAN trunk	Four actuators	Twisted pair; <INSERT gauge>	Verify mating motor-harness pin order
CAN end A / B	CANH–CANL	Bus endpoints	120 Ω each	—	Exactly two terminations after inspection
IMU	SPI, interrupts, 3.3 V, GND	Controller	ISM330DHCX board	Short routed harness/PCB	Quiet return; mechanical orientation label
Receiver	5 V, GND, CRSF TX/RX	Controller/power	RadioMaster RP2 V2	<INSERT>	Verify logic compatibility and failsafe
Emergency isolation	<INSERT>	Battery	Actuator-power branches	Rated device	Reachable outside fall envelope

Chapter D

CAN AND ACTUATOR CONFIGURATION

All entries in this appendix must be read from the installed actuator configuration and released firmware. Manufacturer capability is not equivalent to the chosen project setting. Node identity should be tied to a physical wheel label, connector, wheel coordinate, roller sign, direction multiplier, and encoder zero.

Table D.1. CAN network and actuator-configuration record

Wheel	Physical location y_i	Roller sign s_i	CAN ID	Direction	Mode	Limits	Status
1	<INSERT m>	<INSERT ± 1 >	69 observed	<INSERT>	<VERIFY>	<INSERT>	Partially observed
2	<INSERT m>	<INSERT ± 1 >	<INSERT>	<INSERT>	<VERIFY>	<INSERT>	Pending
3	<INSERT m>	<INSERT ± 1 >	<INSERT>	<INSERT>	<VERIFY>	<INSERT>	Pending
4	<INSERT m>	<INSERT ± 1 >	<INSERT>	<INSERT>	<VERIFY>	<INSERT>	Pending

Table D.2. CAN message-definition record

Frame ID	Direction	DLC	Byte/bit definition	Scale/offset/unit	Period/timeout	Source
<INSERT>	MCU→motor	<INSERT>	Command/mode fields	<INSERT>	20 ms command period observed; verify	Released protocol/firmware
<INSERT>	Motor→MCU	<INSERT>	Position, velocity, torque/current	<INSERT>	<INSERT>	Released protocol/firmware
<INSERT>	Motor→MCU	<INSERT>	Temperature, voltage, fault flags	<INSERT>	<INSERT>	Released protocol/firmware

Note. Include endianness, signedness, saturation behavior, invalid codes, rolling counters, and firmware revision.

Chapter E
FIRMWARE MODULE SUMMARY

Table E.1. Firmware module and verification summary

Layer/module	Representative path	Responsibility	Evidence/status	Required verification
Portable interface	04_firmware/common/include/interfaces/if_spi.h	Abstract SPI transaction contract	Accessible path; implemented interface	Unit/integration error paths
IMU driver	common/include/drivers/drv_ism330dhcx.h; common/src/drivers/drv_ism330dhcx.c	Identity/configuration/raw burst acquisition	Observed code	Register readback and final rate/range
Balance types	common/include/control/ctrl_balance_types.h	Shared balance state/types	Observed scaffolding	Trace to executed controller
Actuator config	platforms/esp32/main/config/config_actuator.h	Node slots and command timing	Four slots; 20 ms observed; one ID observed	All IDs/modes/limits/directions
ESP32 BSP/tasks	04_firmware/platforms/esp32	SPI, TWAI, UART, display, scheduling	Active development implementation	Release commit and worst-case timing
STM32 target	04_firmware/platforms/stm32	Future target/scaffold	Not equivalent to active ESP32	Complete port and regression comparison
Legacy archive	99_archive/documentation_actuator_CubeMars_AK40-10_KV170 and archived generated files	Historical component/code context	Archive only	Do not cite as current without trace
Logging	<PATH NOT CONFIRMED>	Synchronized experiment records	Absent/unverified	Implement schema, time base, dropped-record count
Wheel allocation	<PATH NOT CONFIRMED>	Map body wrench to four commands	Absent/unverified	Implement from measured H with constraints
Balance control	<PATH NOT CONFIRMED>	Closed-loop upright stabilization	Execution not confirmed	Controller, gains, tests, saturation/anti-windup

Recommended release evidence consists of a tagged commit, compiler/toolchain versions, dependency lock or manifest, generated configuration, parameter file, clean-build transcript, binary checksum, flash procedure, hardware revision, and automated test output. Short source excerpts should be added to the thesis only when they clarify an interface or safety decision better than pseudocode.

Chapter F

EXTENDED MODELING DERIVATIONS

F.1 Rank of the Collinear Kinematic Matrix

For the adopted idealized row $h_i = [1, s_i, -y_i]$, the first column is constant, the second represents roller handedness, and the third represents yaw leverage. Rank three requires these three column vectors to be linearly independent. If every s_i is identical, the first two columns are dependent and longitudinal and lateral components cannot be separated. If y is an affine function of s , the third column lies in the span of the first two. Thus a sufficient and readily checked condition is that the roller signs include both handednesses and that the axial coordinates do not collapse into only the affine pattern $y_i = a + b s_i$. The singular values quantify how close the actual geometry is to loss of rank.

F.2 Virtual Work and Wrench Mapping

Equation (6.5) follows from conservation of instantaneous power. With ideal gearing and no roller loss, wheel power is $\tau^T \omega$ and body power is $w^T \nu$. Substituting $\omega = (1/r)H\nu$ and requiring equality for arbitrary ν gives $w = (1/r)H^T \tau$. In a real robot, gear efficiency, current control, roller deformation, and slip modify this relation; experimental identification should therefore compare commanded and measured effort.

F.3 Reduced Linearization

At $\theta = 0$ and zero velocity, $\sin \theta \approx \theta$ and $\cos \theta \approx 1$. Let E be the constant two-by-two mass matrix from Equation (6.8), $D = \text{diag}(b_x, b_\theta)$, and K contain the gravitational term in the pitch row. Then $[\ddot{x}, \ddot{\theta}]^T = E^{-1}(K[x, \theta]^T - D[\dot{x}, \dot{\theta}]^T + B_u u)$. Stacking position and velocity yields Equation (6.9). Signs must be regenerated from the final definition of positive pitch and positive aggregate wheel torque; memorized cart-pole matrices are not acceptable substitutes.

F.4 Allocation under Bounds

Equation (6.6) is the minimum weighted-norm solution when the desired wrench is feasible and no bounds are active. The physical allocation is a constrained quadratic program that minimizes wrench error and actuator effort subject to torque, current, speed, voltage, thermal, and slew bounds. Priority or weighting should preserve the balance-axis command when full planar tracking would consume the available stabilization authority. The allocation residual and active constraints must be logged.

Chapter G
CALIBRATION AND TEST PROTOCOLS

Table G.1. Calibration and experimental protocol record

Protocol	Prerequisites	Procedure summary	Recorded outputs	Acceptance/stop rule
G-01 as-built survey	Power removed; calibrated tools	Label wheels; measure coordinates, radii, mass, COM	Photos, CAD comparison, uncertainties	All parameters traceable
G-02 electrical inspection	Approved schematic and harness	Polarity, resistance, fuse, isolation, termination	Signed checklist/DMM values	No short/reversal; expected bus resistance
G-03 rail bring-up	Current-limited source	Energize logic then actuator branches	V, I, ripple, temperature	Within approved limits; abort on anomaly
G-04 IMU calibration	Final rigid mount	Thermal stationary and six-position records	Bias, scale, noise, transform	Residual within approved limits
G-05 actuator sign test	Wheels unloaded	Command each node at low level	Direction, speed, current, faults	Matches model; immediate disable otherwise
G-06 CAN validation	Two terminations; analyzer	Enumerate nodes; load/timeout/error test	Frames, ages, errors, utilization	No unresolved ID collision/error state
G-07 restrained balance	Fixture, low limits, observer	Verify polarity; raise gains in logged steps	Pitch, commands, states, timing	Stable bounded response/no contact
G-08 free balance	G-07 passed; padded area	Repeated neutral reference trials	All synchronized signals	Approved error/settling and no unsafe event
G-09 disturbance	Free balance passed	Repeat defined input/initial angle	Input proxy, recovery classification	Report boundary; abort on limit
G-10 planar motion	Balance margin known; external pose	Separate then combined reference profiles	Pose, twist, pitch, effort	Approved tracking/cross-coupling limits
G-11 endurance/fault	Thermal/power instrumentation	Repeat duty cycle; inject safe faults	V, I, Wh, temperatures, state	Safe transitions; limits not exceeded

Each protocol requires operator, observer, date, environmental conditions, hardware revision, firmware commit, parameter checksum, battery ID/state, raw-file names, deviations, unsuccessful attempts, and disposition. Approval signatures and institutional laboratory requirements should be added where applicable.

Chapter H

RISK ASSESSMENT

Table H.1 is preliminary and requires review by the author, advisor, and responsible laboratory personnel. Severity and likelihood categories must be replaced by the institution's approved method. It does not supersede battery, machine-shop, electrical, or laboratory policy.

Table H.1. Preliminary risk assessment

Hazard	Cause	Potential consequence	Existing/proposed controls	Residual risk/action
Robot fall/ejection	Unstable control, sign error, saturation	Impact, equipment damage, injury	Restraint, padded exclusion zone, low limits, observer, fall state	Advisor/lab approval required
Wheel pinch/entanglement	Four exposed driven wheels	Pinch, abrasion, entanglement	Guards, tied hair/clothing, remote enable, hands clear	Verify guards and procedure
Unexpected actuation	Startup state, stale command, CAN fault	Motion during handling	Hardware isolation, DISARMED boot, neutral arming, feedback timeout	Validate injected faults
LiPo fire	Overcharge, puncture, short, overcurrent	Fire, toxic smoke, burns	Approved charger/storage, inspection, fuse, containment, supervision	Document exact pack and policy
High fault current	Battery/PCB/harness short	Arcing, conductor heating, fire	Branch fuses, rated connectors/wire, covered terminals, current-limited bring-up	Complete protection design
Hot surfaces	Motor/regulator/trace losses	Burn or component damage	Temperature logging, shutdown limits, cooling interval	Establish measured limits
EMI/communication loss	Motor switching, topology, grounding	Stale feedback and unstable response	Linear CAN, termination, filtering, freshness gate	Bus analyzer and fault test
Structural failure	Mount/shaft/fastener overload	Released component or fall	Inspection, locking hardware, load analysis, guarded test	Confirm loads/materials/safety factor
Software configuration error	Wrong node, gain, units, branch	Uncontrolled response	Versioned configuration, checksums, peer review, staged limits	Add release checklist

Chapter I

REQUIRED FIGURES AND TABLES CHECKLIST

All figure placeholders in the thesis include a proposed number, caption, content description, and likely source. Table I.1 consolidates their production route. Every completed image should be at publication resolution, use legible labels, identify whether it is original or adapted, and preserve editable source files.

Table I.1. Required figure checklist

Figure	Caption subject	How to create/obtain	Status	Required review
1.1	Zephyr robot concept and principal degrees of freedom	Prototype photograph/CAD and thesis workflow diagram	Pending	Author/advisor review
1.2	Research and development workflow used for the Zephyr project	Prototype photograph/CAD and thesis workflow diagram	Pending	Author/advisor review
2.1	Relationship among balancing, omnidirectional mobility, state estimation, and embedded implementation	Original literature-synthesis diagram with citations	Pending	Author/advisor review
3.1	System-level architecture of the Zephyr robot	Original architecture/control/firmware diagram from released design	Pending	Author/advisor review
3.2	Power, sensing, command, and feedback paths among major subsystems	Original architecture/control/firmware diagram from released design	Pending	Author/advisor review
4.1	Current Zephyr mechanical prototype with major components identified	As-built photographs and dimensioned CAD	Pending	Author/advisor review
4.2	Exploded CAD view of the chassis, actuator mounts, wheels, and electronics structure	As-built photographs and dimensioned CAD	Pending	Author/advisor review
4.3	Body coordinate frame, collinear wheel locations, roller signs, and balancing axis	As-built photographs and dimensioned CAD	Pending	Author/advisor review
4.4	Actuator-to-wheel structural interface and load path	As-built photographs and dimensioned CAD	Pending	Author/advisor review
5.1	Electrical architecture and voltage domains	Final KiCad schematic/layout plus annotated wiring photograph	Pending	Author/advisor review
5.2	Battery and high-current power-distribution schematic	Final KiCad schematic/layout plus annotated wiring photograph	Pending	Author/advisor review
5.3	Controller, sensor, receiver, display, and CAN interface schematic	Final KiCad schematic/layout plus annotated wiring photograph	Pending	Author/advisor review
5.4	Recommended linear CAN topology and termination locations	Final KiCad schematic/layout plus annotated wiring photograph	Pending	Author/advisor review
5.5	Final KiCad schematic sheet set	Final KiCad schematic/layout plus annotated wiring photograph	Pending	Author/advisor review
5.6	Final printed-circuit-board layout with power and signal routing identified	Final KiCad schematic/layout plus annotated wiring photograph	Pending	Author/advisor review
6.1	Inertial and body coordinate frames used by the model	Original coordinate/free-body vector drawing from measured geometry	Pending	Author/advisor review
6.2	Idealized force direction for each collinear Mecanum wheel	Original coordinate/free-body vector drawing from measured geometry	Pending	Author/advisor review
6.3	Reduced inverted-pendulum free-body representation	Original coordinate/free-body vector drawing from measured geometry	Pending	Author/advisor review
7.1	Proposed cascaded control and allocation architecture	Original architecture/control/firmware diagram from released design	Pending	Author/advisor review
7.2	Attitude-estimation signal path from raw IMU data to balance state	Original architecture/control/firmware diagram from released design	Pending	Author/advisor review
7.3	Safety-oriented operating-state transitions	Original architecture/control/firmware diagram from released design	Pending	Author/advisor review
8.1	Firmware directory and portability layers	Original architecture/control/firmware diagram from released design	Pending	Author/advisor review

Figure	Caption subject	How to create/obtain	Status	Required review
8.2	Runtime data flow through sensing, estimation, supervision, control, and actuation	Original architecture/control/firmware diagram from released design	Pending	Author/advisor review
8.3	Firmware initialization and arming sequence	Original architecture/control/firmware diagram from released design	Pending	Author/advisor review
8.4	Control-cycle timing and communication schedule	Original architecture/control/firmware diagram from released design	Pending	Author/advisor review
8.5	Motor-command safety gate and fault response	Original architecture/control/firmware diagram from released design	Pending	Author/advisor review
9.1	Simulation model and numerical-integration workflow	Reproducible MATLAB output using final parameters	Pending	Author/advisor review
9.2	Three-dimensional visualization of the collinear Mecanum robot model	Reproducible MATLAB output using final parameters	Pending	Author/advisor review
9.3	Kinematic validation for independent longitudinal, lateral, and yaw commands	Reproducible MATLAB output using final parameters	Pending	Author/advisor review
9.4	Simulated upright-balance response for the selected controller	Reproducible MATLAB output using final parameters	Pending	Author/advisor review
9.5	Sensitivity of balance response to mass properties, delay, and actuator limits	Reproducible MATLAB output using final parameters	Pending	Author/advisor review
10.1	Integrated prototype prepared for controlled bench testing	Dated laboratory photograph/experimental schematic	Pending	Author/advisor review
10.2	Current-limited electrical bring-up and CAN validation setup	Dated laboratory photograph/experimental schematic	Pending	Author/advisor review
10.3	Mechanical restraint used for initial balance-controller tuning	Dated laboratory photograph/experimental schematic	Pending	Author/advisor review
10.4	Synchronized data-acquisition and reference-measurement arrangement	Dated laboratory photograph/experimental schematic	Pending	Author/advisor review
11.1	Measured pitch response during upright balancing	Version-controlled plot from synchronized raw experimental data	Pending	Author/advisor review
11.2	Measured recovery from repeatable pitch disturbances	Version-controlled plot from synchronized raw experimental data	Pending	Author/advisor review
11.3	Measured longitudinal, lateral, and yaw responses	Version-controlled plot from synchronized raw experimental data	Pending	Author/advisor review
11.4	Motor current, battery voltage, and actuator temperature during testing	Version-controlled plot from synchronized raw experimental data	Pending	Author/advisor review
11.5	Measured control-loop timing and CAN communication performance	Version-controlled plot from synchronized raw experimental data	Pending	Author/advisor review

Table I.2. Required table checklist

Table	Title	How to complete	Status	Quality check
1.1	Research questions, objectives, and verification evidence	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
2.1	Comparison of relevant mobile-robot architectures and methods	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
3.1	System requirements and verification matrix	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
3.2	Major subsystem interfaces	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
4.1	Mechanical parameters required for analysis and validation	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
4.2	Mechanical component and interface status	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
5.1	Electrical components and implementation status	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
5.2	Electrical power budget	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
5.3	ESP32 prototype pin assignments	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
5.4	Recommended printed-circuit-board routing classes	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
6.1	Mobility-rank conditions for the collinear wheel arrangement	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
6.2	Kinematic and dynamic symbols	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
6.3	Model parameters and identification method	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
7.1	Control and estimation functions and current status	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
7.2	Controller and estimator parameters requiring final values	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
8.1	Runtime tasks and timing requirements	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
8.2	Proposed operating and safety states	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
8.3	Firmware modules and repository evidence	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
9.1	Simulation cases and required outputs	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
9.2	Simulation parameter record	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
10.1	Integration and experimental test matrix	Populate from approved protocol and raw test data	Ready-fill or draft	Check units, source, and revision
10.2	Test instrumentation and measurement channels	Populate from approved protocol and raw test data	Ready-fill or draft	Check units, source, and revision
10.3	Calibration record	Populate from approved protocol and raw test data	Ready-fill or draft	Check units, source, and revision
11.1	Balancing and motion-performance results	Populate from approved protocol and raw test data	Ready-fill or draft	Check units, source, and revision
11.2	Requirement-verification results	Populate from approved protocol and raw test data	Ready-fill or draft	Check units, source, and revision

Table	Title	How to complete	Status	Quality check
11.3	Failure and anomaly log	Populate from approved protocol and raw test data	Ready-fill or draft	Check units, source, and revision
12.1	Objective completion at the time of this draft	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
A.1	Master system-parameter record	Populate from as-built measurement/configuration	Ready-fill or draft	Check units, source, and revision
B.1	Project evidence and completeness inventory	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
B.2	Proposed thesis titles	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
C.1	ESP32 prototype pin and interface record	Populate from as-built measurement/configuration	Ready-fill or draft	Check units, source, and revision
C.2	Power and CAN connector record	Populate from as-built measurement/configuration	Ready-fill or draft	Check units, source, and revision
D.1	CAN network and actuator-configuration record	Populate from as-built measurement/configuration	Ready-fill or draft	Check units, source, and revision
D.2	CAN message-definition record	Populate from as-built measurement/configuration	Ready-fill or draft	Check units, source, and revision
E.1	Firmware module and verification summary	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
G.1	Calibration and experimental protocol record	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
H.1	Preliminary risk assessment	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
I.1	Required figure checklist	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
I.2	Required table checklist	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
J.1	Consolidated items needed from the author	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
K.1	Reference-verification matrix	Recheck publisher/manufacture landing pages before submission	Ready-fill or draft	Check units, source, and revision
L.1	Final consistency and compliance review	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision
M.1	Repository map and access record	Review repository trace, calculations, and author input	Ready-fill or draft	Check units, source, and revision

Chapter J

ITEMS NEEDED FROM THE AUTHOR

Priority 1 items block defensible modeling, safety review, or experimental claims. Priority 2 items are needed for a complete thesis but do not prevent continued drafting. Priority 3 items improve presentation and archival quality.

Table J.1. Consolidated items needed from the author

Priority	Chapter(s)	Item required	Why needed	Preferred evidence
1	Front matter	Official degree/program, deadline month/year, committee names/titles, copyright year	Template completion	University/advisor confirmation
1	3–6	Final wheel order, roller handedness, wheel coordinates, balancing axis and sign conventions	Determines Jacobian rank and command signs	Annotated as-built photo plus CAD/dimensions
1	4, 6, 9	Total/body mass, COM, inertias, loaded radius, frame geometry	Numerical dynamics and simulation	Measured record and CAD report with uncertainty
1	5, 10	Exact battery label/specification, protection, fuses, wire/connectors, final schematic/PCB	Electrical safety and power claims	Photographs, datasheets, KiCad release
1	5, 8, 10	Installed controller/IMU/CAN hardware revisions and final pin map	Reproducible interface record	Board labels, BOM, released configuration
1	7, 8, 10	Four actuator IDs, modes, direction signs, firmware, limits, scaling, CAN definitions	Safe command generation	Configuration dump and bench log
1	7–11	Executed controller code, estimator/controller rates, gains, filters, limits, state machine	Distinguishes plan from implementation	Tagged commit and parameter file
1	9	Final simulation scripts, parameters, solver settings, and raw outputs	Reproducible simulation results	05_simulation ZIP/release and README
1	10–12	Raw synchronized tests for balance, disturbance, x/y/yaw, timing, power, thermal, faults	Quantitative conclusions	Machine-readable logs plus configuration manifests
1	10	Approved risk assessment, E-stop/power isolation, test-area procedure	Safe experimental authorization	Signed laboratory document
2	1, 4	Design rationale for collinear layout versus conventional four-corner Mecanum	Motivation and tradeoff analysis	Author narrative and dated design notes
2	4	CAD renders, materials, bearings, shafts, couplings, fasteners, assembly torque	Mechanical reproducibility	CAD/BOM/drawings
2	5	PCB copper weight/stack-up, trace/via design, protection selection, fabrication/bring-up results	Electrical verification	Fabrication package and bench logs
2	All	Final repository commit, all branches/tags, full history and archived documentation	Traceability	Git bundle or ZIP plus git log --all
2	11	Approved numerical performance requirements and acceptance thresholds	Pass/fail conclusions	Advisor-approved requirement baseline
2	Front/12	Acknowledgments, funding, disclosures, desired thesis title	Submission completeness	Author text/confirmation
3	All	Publication-quality photographs, diagrams, plots, editable originals and permissions	Presentation quality	CAD/KiCad/MATLAB/vector sources
3	Appendices	Final BOM with costs/vendor revisions and calibration certificates	Reproducibility and archive	Spreadsheet/invoices/certificates

Chapter K

REFERENCE VERIFICATION TABLE

The table records the source identity checked during drafting. Bibliographic metadata should be rechecked against the publisher or manufacturer page immediately before submission because living documentation and datasheet revisions can change. Repository citation [9] must be pinned to the final commit in the submitted version.

Table K.1. Reference-verification matrix

Ref.	Source type	DOI or stable URL	Claims supported
1	Project repository	https://github.com/ahkar-kyaw/e_me599_Zephyr	Implementation paths, project configuration/status
2	Journal article	10.1109/41.982254	Mobile inverted-pendulum precedent
3	Journal article	10.1109/TRO.2004.840905	Wheeled inverted-pendulum control
4	Journal article	10.1109/70.481750	Wheeled-robot kinematic classification
5	Journal article	10.1109/70.313098	Omnidirectional/holonomic platforms
6	Patent	https://patents.google.com/patent/US3876255A/en	Mecanum-wheel origin
7	Journal article	10.1016/j.cagd.2008.07.008	Mecanum-wheel geometry/kinematics
8	Journal article	10.1109/TRO.2008.2010360	Swedish-wheel kinematics/control
9	Conference paper	10.1109/ROBOT.1996.509271	Omniwheel design
10	Journal article	10.1002/rob.4620040209	General WMR kinematic modeling
11	Journal article	10.1016/j.mechatronics.2020.102467	Mecanum roller contact forces
12	Review article	10.1007/s40430-026-06505-4	Current Mecanum modeling/control review
13	Journal article	10.1007/s11768-025-00269-3	Omnidirectional self-balancing control
14	Journal article	10.1109/TAC.2008.923738	Nonlinear complementary attitude filters
15	Textbook	https://fbsbook.org/	Feedback and stability concepts
16	Textbook	ISBN 978-0-486-45766-6 / 1990 edition cited	LQR/state-feedback theory
17	Textbook	ISBN 978-0-13-067389-3	Nonlinear stability/control concepts
18	Journal article	10.1145/321738.321743	Real-time scheduling foundation
19	Official datasheet	https://documentation.espressif.com/esp32_datasheet_en.pdf	ESP32 capabilities/limits
20	Official documentation	https://docs.espressif.com/projects/esp-idf/en/latest/api-reference/peripherals/twai.html	TWAI operation and error states
21	Official datasheet	https://www.st.com/resource/en/datasheet/ism330d-hcx.pdf	IMU registers, rates, ranges
22	Manufacturer page	https://www.cubemars.com/product/ak40-10-robotic-actuator.html	AK40-10 nominal specifications
23	Official datasheet	https://www.nxp.com/docs/en/data-sheet/TJA1051.pdf	CAN transceiver limits/features
24	Official datasheet	https://www.st.com/resource/en/datasheet/stm32h723zg.pdf	Future STM32 target capabilities
25	Manufacturer page	https://www.gobilda.com/gripforce-mecanum-wheel-set-o104mm-40a-durometer-rollers/	Selected wheel nominal specifications
26	Official guide	https://cdn-learn.adafruit.com/downloads/pdf/adafruit-can-pal.pdf	CAN Pal interface
27	Official source	https://github.com/ExpressLRS/ExpressLRS/blob/master/src/include/crsf_protocol.h	CRSF field/protocol definitions
28	Official datasheet	https://www.analog.com/media/en/technical-documentation/data-sheets/LT8609S.pdf	Regulator limits and design basis
29	Official datasheet	https://www.ti.com/lit/ds/symlink/bq76942.pdf	Candidate BMS/protector capabilities
30	Industry standard	IPC-2152	PCB current-carrying design method
31	Official reference design	https://www.ti.com/lit/pdf/tiducf3	Linear CAN topology and termination

Ref.	Source type	DOI or stable URL	Claims supported
32	Official documentation	https://www.freertos.org/Documentation/02-Kernel/02-Kernel-features/01-Tasks-and-co-routines/04-Task-scheduling	RTOS scheduling behavior
33	Manufacturer manuals	https://radiomasterrc.com/pages/user-manuals	Receiver documentation route
34	Manufacturer wiki	https://www.waveshare.com/wiki/2inch_LCD_Module	Display interface documentation
35	Journal article	10.1007/s40435-022-01079-0	Self-balancing trajectory-control comparison

Chapter L

CONSISTENCY AND COMPLIANCE REPORT

This report describes the first-draft state, not final university approval. Page numbers, lists, and pagination were regenerated after document layout; they should be refreshed again after the author replaces placeholders.

Table L.1. Final consistency and compliance review

Check	Draft finding	Remaining action	Owner/review
Template compliance	Uses supplied Fall 2026 template as base; Letter size; 1-in margins; Times New Roman 12 pt body; double-spaced body; required front matter/order; chapter starts	Insert official front-matter data; final university/advisor format review	Author/graduate coordinator
Objective-to-conclusion alignment	Conclusions limited to architecture, subsystem evidence, and symbolic analysis; performance claims withheld	Populate results and revise abstract/conclusions only from verified tests	Author/advisor
Requirement-to-test traceability	Tables 3.1, 10.1, and 11.2 map draft requirements to methods/results	Approve quantitative criteria and execute protocols	Author/advisor
Symbols and units	SI system used; primary symbols listed; pitch/yaw and wheel signs parameterized	Freeze coordinate/sign convention and add measured uncertainties	Author
Figure/table numbering	Chapter-based numbering and lists generated consistently; appendix tables lettered	Refresh after insertions and verify no split captions	Author/editor
Equation numbering	Chapter-based numbering; symbols defined near use or in symbol list	Recheck signs/dimensions with measured geometry; add appendix equations if expanded	Author/advisor
Citation/reference matching	References [9]– [15] are cited and each citation has an entry; primary/official sources prioritized	Reverify final revisions, page ranges, and repository commit	Author/librarian
Unsupported claims	Implementation, plan, and test status explicitly separated; quantitative claims use placeholders	Audit any author replacement text and final captions	Author/advisor
Remaining placeholders	Front matter, hardware measurements, design rationale, figures, numerical parameters, gains, simulation outputs, experiments, results	Resolve Appendix J in priority order	Author
Advisor review	Novelty framing is cautious; collinear rank and balance/control interaction are central	Review thesis scope, theoretical simplifications, requirements, safety, and conclusion strength	Committee
Accessibility	Black text, descriptive captions, table headers, and equation alternative descriptions included	Run final accessibility checker; add alt text to inserted figures	Author/editor
Repository reproducibility	Paths cited where evidence was available; access limitations recorded	Archive final repo, history, raw data, and release manifest	Author

Chapter M

REPOSITORY MAP

Access record for this draft: the public repository landing page, README, root directory names, main-branch status, and earlier inspected project files at commit 97e5a0a were available as evidence. At the current drafting session, GitHub reported 54 commits on the visible main branch and exposed the root locations listed below. Complete recursive traversal of all current files, branches, and history could not be completed through the connected interface, and direct terminal cloning returned an access error. Therefore, current paths not individually reopened are treated as provisional, and no inaccessible file is represented as inspected.

Table M.1. Repository map and access record

Path/location	Observed purpose	Access/evidence state	Needed for final review
README.md	Project description and high-level status	Root item accessible	Pin content to final commit
03_electrical/	KiCad project, schematic, PCB, libraries and controller sheet	Root/folder identity accessible; detailed earlier inspection available	Upload complete current folder including fabrication outputs
04_firmware/common/	Portable interfaces, drivers, control data types	Detailed earlier path/code inspection available	Upload current tree and build/test output
04_firmware/platforms/esp32/	Active ESP32 implementation/configuration	Detailed earlier inspection available	Upload current source, sdkconfig, PlatformIO/ESP-IDF manifests
04_firmware/platforms/stm32/	Future/partial STM32 target	Earlier scaffold evidence	Upload current target and generated configuration
05_simulation/	Simulation material	Visible at current root; contents not recursively accessed	Upload entire folder, dependencies, raw outputs, README
99_archive/documentation_actuator_CubeMars_AK40-10_KV170/Mechanical CAD/location	Archived actuator documentation Frame, mounts, assemblies, drawings	Visible at current root No confirmed current path in accessible root	Upload exact archived documents and identify authoritative revision Upload native CAD, neutral exports, drawings, mass-properties report
Experimental data/location	Raw synchronized validation logs	No confirmed package located	Upload immutable raw data, processing scripts, manifests
Git history/branches/tags	Design evolution and release trace	Landing page reports 54 commits; full traversal incomplete	Provide git bundle or ‘git log --all --decorate --stat’ plus branch/tag archive

Minimum upload requested for a branch-complete evidence audit: a ZIP containing 03_electrical, 04_firmware, 05_simulation, 99_archive, every mechanical/CAD directory, every test-data directory, the README and license, submodules or lockfiles, and a text export of all branch/tag history. A ‘git bundle --all’ is preferable when permitted because it preserves commits, branches, and tags. Sensitive credentials, private keys, and unrelated personal data must be removed before upload.